

The Horizon Structure of Contagion: High-Dimensional Measurement and Inference

BY

Abderrahim Taamouti Endong Wang* Siao Xu[†]

This version: September 25, 2026

ABSTRACT. This paper develops a framework for measuring and conducting inference on horizon-specific contagion networks in high-dimensional restricted predictive systems. We propose a contagion measure that combines statistical evidence of directional predictive transmission with a scale-free measure of its magnitude. For a given source, receiver, and horizon, contagion strength is defined as the log ratio of forecast-error variances with and without the source’s predictive contribution, conditional on the remaining system; edge weights retain this strength only when the corresponding horizon-specific Granger-noncausality hypothesis is rejected. We develop estimation and inference for a broad class of restricted predictive regressions, including heterogeneous autoregressive and grouped-index specifications. Since aggregation and multi-horizon propagation may fail to preserve sparsity, we impose regularity conditions on the grouped predictive state and construct regularized estimators for the high-dimensional regression and covariance objects underlying the relevant forecast-error variances. Applications to large U.S. equity systems document substantial heterogeneity in the frequency and magnitude of predictive transmission across horizons, revealing features of financial connectedness obscured by horizon-agnostic measures.

Keywords: Contagion networks; Granger causality; High-dimensional inference; Systemic risk.

JEL Classification: C32, C55, C58, G01, G10, G17.

¹Department of Economics, University of Mannheim, L7, 3–5, Room 124, 68161 Mannheim, Germany. Email: endong.wang@uni-mannheim.de. Web: <http://www.endongwang.com>.

²Department of Economics, University of Mannheim, L7, 3–5, Room 104, 68161 Mannheim, Germany. Email: siao.xu@students.uni-mannheim.de.

1. INTRODUCTION

A recurring question in the empirical analysis of financial stress is which parts of a system transmit, which absorb, how strongly, and over what horizon. The answer shapes how a network is read. Identifying the sectors that lead the rest of the market at short horizons is relevant for monitoring and for the design of margin and liquidity requirements, whereas identifying those that lead at quarterly horizons speaks to slower balance-sheet and portfolio adjustment. The two need not coincide. A sector may lead others over a few days through order flow and dealer inventories while playing no role at longer horizons, or matter only over a quarter if transmission works through refinancing and credit conditions. A measure of contagion that is not indexed by the forecast horizon averages these channels together.

Measurement also has to answer two questions at once. The first is whether the history of one unit contains information for another beyond what the rest of the system already provides. The second is how much that information improves the forecast. The two margins can point in opposite directions for the same pair. A link that is statistically detectable may carry negligible predictive content, and a large estimated predictive gain is difficult to interpret if the underlying relation cannot be distinguished from zero. A measure that reports only one margin cannot separate these cases, and network aggregates built from it inherit the ambiguity: a dense network of weak links and a sparse network of strong ones can produce the same summary statistic.

The two dominant empirical approaches each address one margin. Forecast-error variance decomposition methods, originating with [Diebold and Yilmaz \(2014\)](#) and extended to large systems and the frequency domain by, among others, [Demirer, Diebold, Liu, and Yilmaz \(2018\)](#) and [Baruník and Křehlík \(2018\)](#), assign continuous weights to directional connectedness. Those weights answer a shock-attribution question and depend on the orthogonalization or generalized normalization used to define innovations; they do not test whether the history of a source carries incremental predictive information for a receiver conditional on the rest of the system. Granger-causality networks define edges directly through predictive content ([Billio, Getmansky, Lo, and](#)

Pelizzon, 2012; Corsi, Lillo, Pirino, and Trapin, 2018), and high-dimensional VAR methods have made system-wide conditioning feasible (Barigozzi and Brownlees, 2019; Chen, Li, Li, and Linton, 2025; Uematsu and Yamagata, 2025). These networks are support-based: an edge records the presence of a predictive relation, usually through the one-step transition law, without attaching to it a scale-free measure of the associated forecast improvement. Franch, Nocciola, and Vouldis (2024) weight time-respecting paths built from Granger-causal links, but by statistical evidence rather than by predictive gain.

A measure that combines the two margins is available in population. The causality measure of Dufour and Taamouti (2010), evaluated at the horizons of the multi-horizon framework of Dufour and Renault (1998), compares forecast-error variances with and without a source. What is missing is a version of it that can be taken to the systems used in financial applications. These are high-dimensional, so that conditioning on the rest of the system is itself an estimation problem, and they are restricted: the predictive state entering each equation is compressed, by temporal aggregation as in the heterogeneous autoregressive model (Corsi, 2009) or by cross-sectional aggregation into sector indices. Existing high-dimensional multi-horizon inference is developed for the unrestricted sparse VAR and targets the horizon-specific coefficients themselves (Dettaa and Wang, 2026). This paper fills that gap.

Our first contribution is the measure. For a fixed source, receiver and horizon, we define the strength of a directed relation as the logarithm of the ratio between the forecast-error variance obtained after removing the predictive block associated with the source and the variance obtained from the receiver’s full information set. It is nonnegative, invariant to the scale of the series, and comparable across pairs and horizons. We separately record whether the horizon-specific Granger-causal relation is active, and take the edge weight of the contagion network to be the product of the two. In population the factorization is redundant, since an inactive edge has zero predictive gain. Its role is in estimation: the reported weight retains an estimated strength only when the corresponding Granger-noncausality hypothesis is rejected, so that network aggregates

weight statistically supported links by their estimated predictive importance instead of treating every rejection alike. Aggregating these weights across sector pairs delivers directional in-, out- and net-contagion profiles indexed by the horizon.

The second contribution is estimation and inference for this measure in high-dimensional restricted predictive systems. We formulate a general grouped predictive regression in which the predictors in the equation for a receiver are a known linear transformation of the unrestricted lagged state, with the transformation allowed to depend on the receiver. The heterogeneous autoregressive specification and a grouped-index VAR, in which own-group variables enter individually while external groups enter through prespecified indices, are the leading cases. The econometric difficulty is that these restrictions cannot be treated as a simplification of the unrestricted problem. Sparsity of the one-step transition matrices does not transfer to the objects that govern inference here: aggregation mixes coordinates, so neither the projection coefficients on the grouped state nor its precision matrix inherits the sparsity pattern of the underlying dynamics, and the same fill-in occurs as the horizon grows and the implied local projections involve powers of the transition matrix. We therefore do not import a sparsity assumption from an unrestricted representation. Regularity conditions are imposed directly on the grouped predictive state at each horizon, and we construct regularized estimators of the nuisance projections it induces.

The third contribution is conceptual and concerns how the two components are estimated. By the Frisch–Waugh–Lovell theorem, the strength measure is a monotone transformation of a quadratic form in the horizon-specific source coefficients, with the form defined by the second-moment matrix of the source block after partialling out the remaining predictors; when that matrix is positive definite, strength is zero exactly when those coefficients are zero. The two components are thus tied to the same parameter but have different statistical characters. The coefficient vector is regular and admits a Gaussian approximation after debiasing, so edge activation can be settled by a test on coefficients. Strength is a nonnegative functional whose derivative vanishes at the null, so it is locally quadratic there and a plug-in built from debiased coefficients would inherit their squared

estimation error. We therefore estimate the two forecast-error variances directly, from a full and a restricted regularized predictive regression. Detection is carried out on coefficients; magnitude is measured by forecast performance. This division of labor is what allows formal Granger-causality inference to be combined with a continuous connection measure without requiring the latter to inherit a Gaussian approximation from the former.

A conceptual restriction should be stated at the outset. Our object is predictive rather than structural, which distinguishes it from contagion tests based on changes in contemporaneous dependence ([Forbes and Rigobon, 2002](#); [Pesaran and Pick, 2007](#)). Conditioning on the remaining system removes predictive content already contained in other observed units and so limits the extent to which common observed information is attributed to a particular directed link, but it does not identify a causal mechanism. Throughout, contagion refers to directional predictive transmission at a given horizon, and Granger causality is used in this predictive sense.

Two applications to large U.S. equities illustrate the two forms of restriction. The first uses daily realized volatility with the HAR grouping to trace the horizon structure of volatility transmission across firms and sectors. The second uses monthly returns and the grouped-index VAR to build a systematic measure of cross-sector contagion at horizons from one month to one year, which we relate to equity-market tail risk measured by ΔCoVaR . Together they show how the screened edge weights move from individual predictive relations to economically interpretable network aggregates.

The remainder of the paper is organized as follows. Section [2](#) defines the horizon-specific contagion measure and the associated network aggregates. Section [3](#) introduces the grouped predictive regression and its HAR and grouped-index VAR special cases. Section [4](#) develops estimation and inference. The subsequent sections report Monte Carlo evidence and the two empirical applications.

2. THE WEIGHTED HORIZON-SPECIFIC CONTAGION NETWORK

This section defines the network as a population object; estimation and inference are developed in Sections 3 and 4. Let $y_t = (y_{1t}, \dots, y_{dt})'$ collect the d units. For each receiver j , let $\mathcal{J}(t)$ denote a prespecified predictive information set available at time t . The network consists of directed, horizon-specific edges and the network-level measures constructed from them.

2.1. Edge measures

Fix a source i , a receiver j , and a horizon $h \geq 1$. Let $\mathcal{J}_{-i}(t)$ denote the information set obtained from $\mathcal{J}(t)$ by removing the predictive block associated with source i , which may represent an individual series or a grouped predictive component, and let $\sigma^2(y_{j,t+h} \mid \mathcal{J})$ denote the population forecast-error variance from projecting $y_{j,t+h}$ on the predictors contained in $\mathcal{J}$.

DEFINITION 2.1 (Granger-causality measure and edge). The horizon- h Granger-causality measure from source i to receiver j is

$$s_{ij}(h) = \log \left(\frac{\sigma^2(y_{j,t+h} \mid \mathcal{J}_{-i}(t))}{\sigma^2(y_{j,t+h} \mid \mathcal{J}(t))} \right), \quad s_{ii}(h) = 0. \quad (2.1)$$

Source i has *incremental predictive content* for the horizon- h value of receiver j , and the directed edge $i \rightarrow j$ is *active* at horizon h , if $s_{ij}(h) > 0$.

Equation (2.1) is the univariate, horizon- h specialisation of the mean-square causality measure of [Dufour and Taamouti \(2010, Definition 4.1\)](#)—their $C_L(y_i \rightarrow_h y_j \mid I)$, with $I_{XY}(t)$ identified with $\mathcal{J}(t)$ and $I_X(t)$ with $\mathcal{J}_{-i}(t)$ —and hence the multi-horizon generalisation of the linear feed-back measures of [Geweke \(1982, 1984\)](#). Since $\mathcal{J}_{-i}(t) \subseteq \mathcal{J}(t)$, the two projections are nested and $s_{ij}(h) \geq 0$. Being a log ratio of nested forecast-error variances, $s_{ij}(h)$ is approximately the fractional reduction in the horizon- h forecast-error variance of j attributable to the history of i , exactly $1 - e^{-s_{ij}(h)}$; being a ratio, it is invariant to scale and comparable across pairs and across horizons.

Remark 2.1. The measure is defined relative to a prespecified information set rather than through a decomposition of structural innovations, and so requires neither orthogonalisation nor generalised normalisation. It is conditional in the sense of Geweke (1984): the remaining units are held in the conditioning set as auxiliary variables rather than marginalised out.

An edge may also be represented by its presence alone. Let $H_{0,ij}(h) : s_{ij}(h) = 0$ denote the hypothesis of Granger non-causality from i to j at horizon h in the sense of Dufour and Renault (1998), tested against the one-sided alternative $H_{1,ij}(h) : s_{ij}(h) > 0$.

DEFINITION 2.2 (Binary Granger-causality measure). The horizon- h binary Granger-causality measure from source i to receiver j is

$$c_{ij}(h) = \mathbf{1}\{H_{0,ij}(h) \text{ false}\}, \quad c_{ii}(h) = 0. \quad (2.2)$$

The two margins are combined in the contagion weight

$$S_{ij}(h) = c_{ij}(h) s_{ij}(h), \quad (2.3)$$

which assigns to each active edge its predictive strength and zero otherwise. The factors enter for different reasons, and are recovered by different devices: $s_{ij}(h)$ is a parameter and supplies the economic content of an edge, while $c_{ij}(h)$ is what a test of $H_{0,ij}(h)$ delivers and makes $S_{ij}(h)$ a thresholding functional in which the test supplies the threshold; Remark 2.3 below explains why the second factor is needed once edges are aggregated.

Remark 2.2 (Relation to the literature). Existing constructions emphasise one of the two margins in (2.3). Connectedness measures based on forecast-error variance decompositions (Baruník and Křehlík, 2018; Demirer et al., 2018; Diebold and Yilmaz, 2014) deliver continuous directional weights, but these allocate forecast variation across innovations and depend on the identification scheme; being row-normalised shares, they also cannot express a system in which transmission is uniformly weak. Granger-causality networks record incremental predictive content directly,

and high-dimensional VAR methods have made system-wide conditioning feasible (Barigozzi and Brownlees, 2019; Chen et al., 2025; Uematsu and Yamagata, 2025); their output, however, is the support of the network. In Detta and Wang (2026) we develop inference for horizon-specific Granger causality in a high-dimensional VAR, which determines $c_{ij}(h)$ but attaches no strength to an edge. Applied constructions that combine a causality measure with a significance filter (Corsi et al., 2018; Papana, Kyrtsov, Kugiumtzis, and Diks, 2017) discard the magnitude once the support is determined; the present construction retains it.

Varying h traces the *horizon structure* of contagion and shows how both the prevalence and the strength of predictive transmission vary from the short to the long run. Because $s_{ij}(h)$ conditions on the remaining predictors, an edge active at $h > 1$ but inactive at $h = 1$ reflects transmission routed through other units, the indirect causality of Dufour and Renault (1998).

2.2. Network measures

Consider two economically defined groups $\mathcal{G}_a$ and $\mathcal{G}_b$, such as sectors, asset classes, or regions, with $N_a = |\mathcal{G}_a|$ and $N_b = |\mathcal{G}_b|$. For disjoint source and receiver groups, define the group-level weighted Granger-causality index by

$$\text{WGC}_{\mathcal{G}_a \rightarrow \mathcal{G}_b}(h) = \frac{1}{N_a N_b} \sum_{i \in \mathcal{G}_a} \sum_{j \in \mathcal{G}_b} S_{ij}(h). \quad (2.4)$$

Inactive links enter as zeros and active links with their strength, so the index reflects both how many links transmit and how much each transmits, and is interpretable as an average percentage reduction in horizon- h forecast-error variance. Comparing it with its reverse direction identifies whether predictive transmission runs predominantly from $\mathcal{G}_a$ to $\mathcal{G}_b$ or vice versa. The sample counterpart replaces $S_{ij}(h)$ by $\hat{S}_{ij}(h)$ and thus aggregates the predictive importance of statistically supported links rather than treating all links equally.

When the source and receiver sets are the entire system, with self-links excluded, this reduces to $\text{WGC}(h) = [d(d-1)]^{-1} \sum_{i=1}^d \sum_{j \neq i} S_{ij}(h)$, the average weighted transmission in the system-

wide network, whose horizon profile shows where predictive transmission is concentrated. Unlike the variance-decomposition and frequency-based connectedness measures of [Diebold and Yilmaz \(2014\)](#) and [Baruník and Křehlík \(2018\)](#), to which it is the Granger-causality counterpart, $\text{WGC}(h)$ is not normalised to a fixed total and can therefore fall towards zero at horizons at which the system transmits little.

For a group $\mathcal{G}_a$, let $\mathcal{G}_{-a} = \{1, \dots, d\} \setminus \mathcal{G}_a$ and define $\text{WGC}_{\mathcal{G}_a}^{\text{out}}(h) = \text{WGC}_{\mathcal{G}_a \rightarrow \mathcal{G}_{-a}}(h)$ and $\text{WGC}_{\mathcal{G}_a}^{\text{in}}(h) = \text{WGC}_{\mathcal{G}_{-a} \rightarrow \mathcal{G}_a}(h)$. Their difference identifies groups that are net transmitters or net receivers of contagion and shows how these roles vary across horizons; because every edge is defined relative to the remaining predictors, these directional measures capture incremental predictive transmission rather than unconditional comovement. Equal weights keep the interpretation transparent, but predetermined nonnegative weights can be used when the application calls for value or exposure weighting.

Remark 2.3 (Why the factorization). The binary margin is what makes the aggregates above well behaved in high dimensions. Nesting forces $\widehat{s}_{ij}(h) \geq 0$ for every pair, so estimation error on inactive edges is one-signed and does not cancel when averaged over the $d(d-1)$ candidate links; without the indicator this contamination enters (2.4) and its system-wide counterpart directly. The resulting estimator is a hard-thresholding estimator of the type analysed by [Bickel and Levina \(2008\)](#), with a threshold studentised pair-by-pair as in [Cai and Liu \(2011\)](#) and, since network sparsity is unknown and varies with h , determined by the data rather than fixed. Ordering the $N = d(d-1)$ p -values of the tests of $H_{0,ij}(h)$ as $p_{(1)}(h) \leq \dots \leq p_{(N)}(h)$, the step-up rule of [Benjamini and Hochberg \(1995\)](#) at false discovery rate q retains edges below

$$t_q(h) = \frac{qk^*(h)}{N}, \quad k^*(h) = \max\{k : p_{(k)}(h) \leq qk/N\}, \quad (2.5)$$

the level scaled by the realised proportion of active edges, so that the rule tightens automatically as the network becomes sparser, with $t_q(h) \in [q/N, q]$. [Abramovich, Benjamini, Donoho, and Johnstone \(2006\)](#) show that a threshold of this form adapts to unknown sparsity in the Gaussian

sequence model. Section 4 develops the estimator.

3. GROUPED PREDICTIVE REGRESSIONS

3.1. Grouped explanatory variables

Let $y_t = (y_{1t}, \dots, y_{dt})'$ be a d -dimensional centred series. For a lag order p , collect the full predictive state $Y_t := (y'_t, y'_{t-1}, \dots, y'_{t-p+1})' \in \mathbb{R}^{pd}$. A standard multi-horizon predictive regression conditions on Y_t directly. We instead allow the predictive state to be grouped in a manner that depends on the receiver. For each receiver j , let $G_j \in \mathbb{R}^{r_j \times pd}$ be a known linear map and set $X_{j,t} := G_j Y_t \in \mathbb{R}^{r_j}$. The map G_j is fixed across horizons but may vary across receivers, so that different outcome equations may condition on different grouped representations of the same state.

For a source i , let $R_{ij} \in \mathbb{R}^{k_{ij} \times r_j}$ select the block of $X_{j,t}$ associated with i and let $R_{-i,j} \in \mathbb{R}^{(r_j - k_{ij}) \times r_j}$ select the remainder, and write $X_{ij,t} = R_{ij} X_{j,t}$ and $X_{-i,j,t} = R_{-i,j} X_{j,t}$. The horizon- h equation for receiver j is

$$y_{j,t+h} = \beta'_{ij,h} X_{ij,t} + \gamma'_{ij,h} X_{-i,j,t} + u_{j,t,h}. \quad (3.1)$$

Thus $\beta_{ij,h}$ measures the incremental predictive content of the source block of i for $y_{j,t+h}$ after conditioning on the remaining predictors in $X_{j,t}$.

Let $\Sigma_Y := \mathbb{E}[Y_t Y'_t]$ and define the receiver- j state covariance and precision matrices by $\Sigma_{X,j} := \mathbb{E}[X_{j,t} X'_{j,t}] = G_j \Sigma_Y G'_j$ and $\Theta_j := \Sigma_{X,j}^{-1}$. For the source block i , define

$$\Sigma_{ij \cdot -i} = (R_{ij} \Theta_j R'_{ij})^{-1}. \quad (3.2)$$

This is the covariance matrix of the residual from projecting $X_{ij,t}$ on $X_{-i,j,t}$.

Let $\sigma_j^2(h)$ be the residual variance from projecting $y_{j,t+h}$ on $X_{j,t}$ and let $\sigma_{j \setminus i}^2(h)$ be that from

projecting on $X_{-i,j,t}$ alone. By the Frisch–Waugh–Lovell theorem,

$$\sigma_{j \setminus i}^2(h) = \sigma_j^2(h) + \beta'_{ij,h} \Sigma_{ij \cdot -i} \beta_{ij,h}. \quad (3.3)$$

Hence the Granger-causality measure of Definition 2.2 can be written as

$$s_{ij}(h) = \log \left(\frac{\sigma_{j \setminus i}^2(h)}{\sigma_j^2(h)} \right) = \log \left(1 + \frac{\beta'_{ij,h} \Sigma_{ij \cdot -i} \beta_{ij,h}}{\sigma_j^2(h)} \right). \quad (3.4)$$

Provided the population projection is uniquely defined, the horizon- h null of Granger non-causality can equivalently be written as $H_0(i \rightarrow j, h) : s_{ij}(h) = 0$ if and only if $\beta_{ij,h} = 0$. Because both the coefficient and the conditioning state are indexed by the receiver, the relevant information set may differ across outcome equations while remaining fixed across horizons for a given receiver. The sequence $\{s_{ij}(h)\}_{h \geq 1}$ therefore traces the horizon structure of directed predictive transmission from i to j .

3.2. Leading special cases: the HAR and the grouped-index VAR

Two choices of the map organise the analysis. The HAR specification applies the same temporal grouping to every receiver, whereas the grouped-index VAR applies a receiver-group-specific map that compresses the cross-section.

3.2.1. The HAR grouping

The HAR specification of Corsi (2009) and Hecq, Margaritella, and Smeekes (2023) sets $p = 22$ and groups the lag history into daily, weekly and monthly components, $y_t^{(w)} = \frac{1}{5} \sum_{\ell=0}^4 y_{t-\ell}$ and $y_t^{(m)} = \frac{1}{22} \sum_{\ell=0}^{21} y_{t-\ell}$. For every receiver j the grouped state is $X_{j,t} = X_t^{\text{HAR}} := \left(y_t', y_t^{(w)'} , y_t^{(m)'} \right)' \in$

$\mathbb{R}^{3d}$, so that the map is common, $G_j = G^{\text{HAR}}$ for all j , with

$$G^{\text{HAR}} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ \frac{1}{5} & \frac{1}{5} & \cdots & 0 \\ \frac{1}{22} & \frac{1}{22} & \cdots & \frac{1}{22} \end{pmatrix} \otimes I_d \in \mathbb{R}^{3d \times 22d}, \quad (3.5)$$

where the second row places $1/5$ in its first five entries and zero thereafter. The source block $X_{ij,t} = (y_{i,t}, y_{i,t}^{(w)}, y_{i,t}^{(m)})'$ does not depend on j , and $\beta_{ij,h} \in \mathbb{R}^3$ collects the daily, weekly and monthly effects of i on j at horizon h ; correspondingly $\Sigma_{X,j} = \Sigma_X$ and $\Theta_j = \Theta$ for every j . The restriction compresses the lag dimension while retaining all d variables in the cross-section: relative to an unrestricted VAR(22) state it replaces the 22 lag coefficients per variable by three fixed-weight temporal components, chosen to capture the multi-scale persistence of realised volatility.

3.2.2. The grouped-index VAR

The grouped-index VAR compresses the cross-section instead. Partition the d variables into $\bar{g}$ possibly unequal groups $\mathcal{G}_1, \dots, \mathcal{G}_{\bar{g}}$, with $n_g = |\mathcal{G}_g|$ and $\sum_g n_g = d$, and let $g(j)$ denote the group containing receiver j . Let $J_g \in \mathbb{R}^{n_g \times d}$ be the selection matrix for $\mathcal{G}_g$, so that $J_g y_t = (y_{i,t})_{i \in \mathcal{G}_g}$. For each receiver group g and external group $k \neq g$, let $\omega_{gk} \in \mathbb{R}^{n_k}$ satisfy $\mathbf{1}' \omega_{gk} = 1$ and define the external-group index as $y_{gk,t}^* = \omega_{gk}' J_k y_t$.

For $j \in \mathcal{G}_g$, own-group variables enter individually, whereas each external group enters through a single index:

$$y_{j,t+h} = \sum_{\ell=0}^{p-1} \sum_{i \in \mathcal{G}_g} a_{ji,\ell,h} y_{i,t-\ell} + \sum_{\ell=0}^{p-1} \sum_{\substack{k=1 \\ k \neq g}}^{\bar{g}} b_{jk,\ell,h} y_{gk,t-\ell}^* + u_{j,t,h}. \quad (3.6)$$

Receivers in the same group therefore share the same grouped predictive state, while their coefficients remain receiver- and horizon-specific.

In the notation of Section 3.1, for $j \in \mathcal{G}_g$ the corresponding map is

$$G_j = G_g = I_p \otimes \begin{pmatrix} J_g \\ (\omega'_{gk} J_k)_{k \neq g} \end{pmatrix} \in \mathbb{R}^{p(n_g + \bar{g} - 1) \times pd}, \quad (3.7)$$

where each $\omega'_{gk} J_k \in \mathbb{R}^{1 \times d}$ forms the index row for external group k , and $(\omega'_{gk} J_k)_{k \neq g} \in \mathbb{R}^{(\bar{g}-1) \times d}$ stacks these rows. The map retains every own-group variable and replaces each external group by one weighted index at each lag. It is common to receivers within the same group but may differ across groups.

For $i \in \mathcal{G}_{g(j)}$, the source block $X_{ij,t}$ contains $(y_{i,t}, \dots, y_{i,t-p+1})'$, so that $\beta_{ij,h} \in \mathbb{R}^p$ measures the horizon- h predictive effect of i on j . For an external group $k \neq g(j)$, the corresponding source block contains $(y_{g(j)k,t}^*, \dots, y_{g(j)k,t-p+1}^*)'$, with coefficient vector $(b_{jk,0,h}, \dots, b_{jk,p-1,h})'$. Thus predictive transmission is individual-specific within the receiver's own group and group-specific for external groups.

This specification differs from the network autoregression of Zhu, Pan, Li, Liu, and Wang (2017) and its grouped extension (Zhu and Pan, 2020), which restrict cross-sectional response coefficients across nodes or communities. Here the grouping instead restricts the predictive information set: receivers within a group share the same grouped state, while $a_{ji,\ell,h}$ and $b_{jk,\ell,h}$ remain receiver- and horizon-specific.

4. HIGH-DIMENSIONAL INFERENCE

This section develops a debiased regularised estimator of $\beta_{ij,h}$ and an estimator of $s_{ij}(h)$. Throughout, we work on a single estimation window and suppress the window subscript. We retain the receiver-specific notation of Section 3.1, which covers both leading cases: G_j is common across receivers under the HAR grouping and satisfies $G_j = G_{g(j)}$ under the grouped-index VAR.

4.1. Debiased regularised estimator

Recall that $X_{ij,t} = R_{ij}X_{j,t}$ denotes the source block and $X_{-i,j,t} = R_{-i,j}X_{j,t}$ the remaining predictors. The population orthogonalised source block is $v_{ij,t} = X_{ij,t} - \mathbb{E}(X_{ij,t} | X_{-i,j,t}) = (R_{ij}\Theta_j R'_{ij})^{-1} R_{ij}\Theta_j X_{j,t}$, where the conditional expectation denotes the population linear projection. We estimate it by $\hat{v}_{ij,t} = (R_{ij}\hat{\Theta}_j R'_{ij})^{-1} R_{ij}\hat{\Theta}_j X_{j,t}$. Since the orthogonalisation does not depend on h , it is computed once for each source–receiver block and reused across horizons.

Let $b_{j,h}$ denote the coefficient from projecting $y_{j,t+h}$ on $X_{j,t}$. The nuisance coefficient in (3.1) is $\gamma_{ij,h} = R_{-i,j}b_{j,h}$, and we use $\hat{\gamma}_{ij,h} = R_{-i,j}\hat{b}_{j,h}$. The estimators $\hat{\Theta}_j$ and $\hat{b}_{j,h}$ are constructed in Section 4.2.

Write $T := n - h - p + 1$ for the effective sample size. The debiased regularised estimator is

$$\hat{\beta}_{ij,h} = \left(\sum_{t=p}^{n-h} \hat{v}_{ij,t} X'_{ij,t} \right)^{-1} \sum_{t=p}^{n-h} \hat{v}_{ij,t} \left(y_{j,t+h} - \hat{\gamma}'_{ij,h} X_{-i,j,t} \right). \quad (4.1)$$

By construction, $\mathbb{E}[v_{ij,t} X'_{-i,j,t}] = 0$. Moreover, any directional derivative of $v_{ij,t}$ with respect to Θ_j is linear in $X_{j,t}$, so that $\mathbb{E}[(\partial_{\Theta_j} v_{ij,t}) u_{j,t,h}] = 0$ by $\mathbb{E}[X_{j,t} u_{j,t,h}] = 0$. The resulting score is therefore Neyman-orthogonal with respect to the first-step nuisance components (Chernozhukov, Chetverikov, Demirer, Duflo, Hansen, Newey, and Robins, 2018).

For inference, we use the full-projection residual $\hat{u}_{j,t,h} = y_{j,t+h} - \hat{b}'_{j,h} X_{j,t}$. The h -step projection error is generally serially correlated for $h > 1$ and, because $X_{j,t} = G_j Y_t$ is a grouped state, need not be MA($h-1$). Let $\Omega_{ij,h}$ denote the long-run variance of $v_{ij,t} u_{j,t,h}$. We estimate it by the Bartlett Newey–West estimator

$$\hat{\Omega}_{ij,h} = \sum_{|\ell| \leq b_n} \kappa(\ell/b_n) \hat{\Gamma}_\ell, \quad (4.2)$$

where $\hat{\Gamma}_\ell = T^{-1} \sum_t \hat{v}_{ij,t} \hat{u}_{j,t,h} \hat{u}_{j,t-\ell,h} \hat{v}'_{ij,t-\ell}$ and $\kappa(x) = (1 - |x|)_+$. Consistency of the feasible HAC estimator is established under the bandwidth, moment, and growth conditions of Section 4.3. The

asymptotic-variance estimator is

$$\widehat{\text{AVar}}(\hat{\beta}_{ij,h}) = T \left(\sum_{t=p}^{n-h} \hat{v}_{ij,t} X'_{ij,t} \right)^{-1} \hat{\Omega}_{ij,h} \left(\sum_{t=p}^{n-h} \hat{v}_{ij,t} X'_{ij,t} \right)^{t-1}. \quad (4.3)$$

The Wald statistic for $H_0(i \rightarrow j, h)$ is

$$W_{ij,h} = \hat{\beta}'_{ij,h} \left[\widehat{\text{AVar}}(\hat{\beta}_{ij,h}) \right]^{-1} \hat{\beta}_{ij,h}. \quad (4.4)$$

Under the null and the regularity conditions of Section 4.3, $W_{ij,h} \xrightarrow{d} \chi^2_{k_{ij}}$, where $k_{ij} = 3$ under the HAR grouping and $k_{ij} = p$ under the grouped-index VAR. Hence the null is rejected at level α whenever $W_{ij,h} > \chi^2_{k_{ij}, 1-\alpha}$. The theory is pointwise in the source–receiver pair and horizon. For simultaneous network inference, standard FWER or FDR adjustments may instead be applied to the collection of Wald p -values.

Recall that $\Sigma_{ij \cdot -i} = (R_{ij} \Theta_j R'_{ij})^{-1}$. We estimate it by $\hat{\Sigma}_{ij \cdot -i} = (R_{ij} \hat{\Theta}_j R'_{ij})^{-1}$ and estimate the full-projection residual variance by $\hat{\sigma}_j^2(h) = T^{-1} \sum_{t=p}^{n-h} \hat{u}_{j,t,h}^2$. The estimator of the Granger-causality measure is then

$$\hat{s}_{ij}(h) = \log \left(1 + \frac{\hat{\beta}'_{ij,h} \hat{\Sigma}_{ij \cdot -i} \hat{\beta}_{ij,h}}{\hat{\sigma}_j^2(h)} \right). \quad (4.5)$$

Thus estimation of $s_{ij}(h)$ requires no additional restricted high-dimensional regression.

4.2. High-dimensional regularised estimation

We now construct the nuisance estimators used above. The common ingredient is an estimate of the covariance of the full predictive state Y_t . Receiver-specific objects are then obtained by applying the corresponding map G_j .

4.2.1. Precision matrix estimation

For each receiver j , estimate the horizon-one projection on $X_{j,t} = G_j Y_t$ by

$$\tilde{b}_{j,1} = \arg \min_{b \in \mathbb{R}^{r_j}} \frac{1}{n-p} \sum_{t=p}^{n-1} (y_{j,t+1} - b' X_{j,t})^2 + \lambda \|\Pi_j b\|_1. \quad (4.6)$$

Here Π_j is a diagonal penalty-loading matrix. We use a preliminary Lasso selected by a BIC-type criterion, followed by adaptive loadings proportional to $(|\tilde{b}_{j,\ell}^{\text{pre}}| + (n-p)^{-1/2})^{-1}$, with the second-step tuning parameter selected by the same criterion (Zou, 2006; Zou, Hastie, and Tibshirani, 2007).

The receiver-specific maps imply a common restricted VAR(p) representation for y_t . In particular, the j -th row of its coefficient matrix satisfies $[A_1, \dots, A_p]_{j,\cdot} = b'_{j,1} G_j$. We therefore set

$$[\hat{A}_1, \dots, \hat{A}_p]_{j,\cdot} = \text{THR}_{\tau_B}(\tilde{b}_{j,1})' G_j, \quad (4.7)$$

where THR_{τ_B} is applied entrywise and $\tau_B \asymp \sqrt{v_n/n}$. This formulation covers both the HAR case, where the G_j coincide, and the grouped-index VAR, where G_j and r_j may vary across receiver groups.

Let $\hat{\mathbf{A}}$ denote the companion matrix associated with $\hat{A}_1, \dots, \hat{A}_p$, and let $\hat{u}_{t,1} = y_{t+1} - \sum_{\ell=1}^p \hat{A}_\ell y_{t-\ell+1}$. The innovation covariance is estimated by thresholding $(n-p)^{-1} \sum_{t=p}^{n-1} \hat{u}_{t,1} \hat{u}_{t,1}'$, yielding $\hat{\Sigma}_u$.

With $J = [I_d, \mathbf{0}]$, the stationary covariance of the full state is estimated by

$$\hat{\Sigma}_Y = \sum_{m=0}^{N_n} \hat{\mathbf{A}}^m J' \hat{\Sigma}_u J (\hat{\mathbf{A}}')^m. \quad (4.8)$$

The receiver-specific covariance and precision estimators are then $\hat{\Sigma}_{X,j} = G_j \hat{\Sigma}_Y G_j'$ and $\hat{\Theta}_j = \hat{\Sigma}_{X,j}^{-1}$. The truncation level N_n is chosen so that the omitted tail is asymptotically negligible under the growth conditions of Section 4.3; in the numerical implementation we use $N_n = 50$. An eigenvalue floor is applied before inversion when necessary and is asymptotically inactive under the conditions

below.

4.2.2. Projection coefficient estimation

The companion representation implies $\mathbb{E}[X_{j,t}y_{j,t+h}] = G_j \Sigma_Y (\mathbf{A}')^h J' e_j$, where e_j is the j -th canonical basis vector in $\mathbb{R}^d$. Hence the full-state coefficient is estimated by

$$\hat{b}_{j,h} = \hat{\Theta}_j G_j \hat{\Sigma}_Y (\hat{\mathbf{A}}')^h J' e_j. \quad (4.9)$$

Its $-i$ block gives $\hat{\gamma}_{ij,h} = R_{-i,j} \hat{b}_{j,h}$.

The resulting multi-horizon Granger-causality notion is relative to the grouped predictive state $X_{j,t} = G_j Y_t$. The dimension reduction encoded by G_j is therefore part of the maintained predictive specification. The restricted VAR is used to estimate the second moments and cross-moments entering these projections, while Neyman orthogonalisation protects $\hat{\beta}_{ij,h}$ against sufficiently small first-step estimation errors. The required first-step rates and growth conditions are stated in Section 4.3.

4.3. Asymptotic properties

This subsection establishes the asymptotic normality of the debiased estimator $\hat{\beta}_{ij,h}$ and the consistency of $\hat{s}_{ij}(h)$. Throughout, (i, j) and h are fixed, $T/n \rightarrow 1$, and $v_n, \tilde{v}_n \geq 1$ satisfy $v_n/n \rightarrow 0$ and $\tilde{v}_n/n \rightarrow 0$. Under sub-Gaussian tails and fixed p , both may be taken of logarithmic order in d . Following [Bickel and Levina \(2008\)](#), let $\mathcal{U}(k, \mu) = \{M = (m_{ab}) : \max_a \sum_b |m_{ab}|^\mu \leq k\}$ for $\mu \in [0, 1)$.

ASSUMPTION 4.1. The following conditions hold.

- (i) The restricted VAR(p) is stable. For some $\varphi \in (0, 1)$, $\|\mathbf{A}^m\|_2 \leq C\varphi^m$ and $\|\mathbf{A}^m\|_l \leq Ck_A \varphi^m$ for $l \in \{1, \infty\}$ and all $m \geq 0$.

- (ii) For some $\mu \in [0, 1)$ and $k_B \geq 1$, the regularised first-step estimator satisfies $\|[\hat{A}_1, \dots, \hat{A}_p] - [A_1, \dots, A_p]\|_l = O_p(k_B^{1.5}(\mathbf{v}_n/n)^{(1-\mu)/2})$ for $l \in \{1, \infty\}$.
- (iii) The innovation covariance satisfies $\Sigma_u \in \mathcal{U}(k_u, \mu_u)$ for some $\mu_u \in [0, 1)$ and $k_u \geq 1$, $\|\Sigma_u\|_\infty = O(k_u)$, and $\lambda_{\min}(\Sigma_u) \geq c > 0$. The one-step innovation $u_{t,1}$ is a martingale difference sequence. For all deterministic $U \in \mathbb{R}^{q_1 \times d}$ and $V \in \mathbb{R}^{d \times q_2}$ with $q_1, q_2 = O(d)$ and $\|U\|_2 = \|V\|_2 = 1$, $\|n^{-1} \sum_{t=1}^n U(u_{t,1} u'_{t,1} - \Sigma_u) V\|_{\max} = O_p(\sqrt{\tilde{\mathbf{v}}_n/n})$.
- (iv) For each fixed (i, j, h) , $\|T^{-1} \sum_t X_{j,t} u_{j,t,h}\|_{\max} = O_p(\sqrt{\tilde{\mathbf{v}}_n/n})$.
- (v) For each receiver j , there exists $k_\Theta \geq 1$ such that $\|\Theta_j\|_\infty = O(k_\Theta)$.
- (vi) The score $v_{ij,t} u_{j,t,h}$ is α -mixing with coefficients $\{\alpha(\ell)\}_{\ell \geq 1}$ satisfying $\sum_{\ell \geq 1} \ell^{r_0/(r_0-2)-1} \alpha(\ell) < \infty$ for some $r_0 > 2$, and $\mathbb{E}|\lambda' v_{ij,t} u_{j,t,h}|^{4r_0+\delta} \leq C$ for every λ with $\|\lambda\|_1 = 1$. The process $u_{j,t,h}^2$ is also α -mixing with sufficiently high finite moments. The long-run variance $\Omega_{ij,h}$ satisfies $\lambda_{\min}(\Omega_{ij,h}) \geq c > 0$, and $\sigma_j^2(h) \geq c > 0$.

Assumption 4.1(i) controls propagation of estimation error through the stable VAR, while part (ii) imposes the regularised first-step rate. Part (iii) gives the innovation-covariance conditions used to estimate the stationary covariance of the full predictive state. Together with stability, its concentration condition also yields the corresponding entrywise concentration for the grouped-state covariance. Part (iv) is the additional concentration condition required because the grouped-state projection error $u_{j,t,h}$ need not be a finite combination of future innovations. Part (v) controls the receiver-specific precision matrix, and part (vi) provides the weak-dependence conditions used for the score CLT and the residual-variance law of large numbers.

Define

$$\delta_n = k_u k_B^3 (\mathbf{v}_n/n)^{1-\mu} + \sqrt{\tilde{\mathbf{v}}_n/n}, \quad Q_n = k_A k_B^{1.5} (\mathbf{v}_n/n)^{(1-\mu)/2} + \delta_n^{1-\mu_u} + \varphi^{2N_n}. \quad (4.10)$$

The two terms in δ_n correspond to the effect of estimating the one-step dynamics on the innova-

tion covariance and its sampling fluctuation, respectively. The three terms in Q_n correspond to estimation of the dynamic coefficients, estimation of the innovation covariance, and truncation of the stationary covariance representation. We maintain $k_A k_B^{1.5} (v_n/n)^{(1-\mu)/2} = o(1)$, $\delta_n = o(1)$, and $N_n \rightarrow \infty$.

For inference on $\beta_{ij,h}$, suppose

$$\tilde{v}_n^{1/2} k_\Theta^3 k_u^2 k_A^5 Q_n + n^{1/2} k_\Theta^4 k_u^3 k_A^7 Q_n^2 = o(1). \quad (4.11)$$

The first term controls the marginal first-step errors entering the empirical orthogonality conditions, while the second controls their product at the $\sqrt{n}$ scale.

For feasible HAC estimation, let $b_n \rightarrow \infty$ and $b_n/n \rightarrow 0$, suppose $T^{-1} \sum_t \|X_{j,t}\|_\infty^4 = O_p(\tilde{v}_n^2)$, $T^{-1} \sum_t u_{j,t,h}^4 = O_p(1)$, and $T^{-1} \sum_t \|v_{ij,t}\|_2^4 = O_p(1)$, and require

$$b_n \tilde{v}_n^{1/2} k_\Theta^3 k_u^3 k_A^5 Q_n = o(1). \quad (4.12)$$

This condition prevents the first-step error in the feasible score from accumulating over the increasing number of HAC autocovariances.

Recall that $\Sigma_{ij,-i} = \mathbb{E}[v_{ij,t} v'_{ij,t}] = (R_{ij} \Theta_j R'_{ij})^{-1}$.

THEOREM 4.1 (Asymptotic normality). *Under Assumption 4.1, suppose (4.11) and (4.12) hold together with the stated bandwidth and moment conditions. Then*

$$\sqrt{T} (\hat{\beta}_{ij,h} - \beta_{ij,h}) = \Sigma_{ij,-i}^{-1} T^{-1/2} \sum_t v_{ij,t} u_{j,t,h} + o_p(1), \quad (4.13)$$

and

$$\sqrt{T} (\hat{\beta}_{ij,h} - \beta_{ij,h}) \xrightarrow{d} N\left(0, \Sigma_{ij,-i}^{-1} \Omega_{ij,h} \Sigma_{ij,-i}^{-1}\right). \quad (4.14)$$

Moreover, $T \widehat{\text{AVar}}(\hat{\beta}_{ij,h}) \xrightarrow{p} \Sigma_{ij,-i}^{-1} \Omega_{ij,h} \Sigma_{ij,-i}^{-1}$. Consequently, under $H_0(i \rightarrow j, h)$, $W_{ij,h} \xrightarrow{d} \chi_{k_{ij}}^2$.

Under the local alternative $\beta_{ij,h} = T^{-1/2}\bar{\beta}$, the Wald statistic converges to a noncentral $\chi^2_{k_{ij}}$ distribution with noncentrality parameter $\bar{\beta}'\Sigma_{ij\cdot-i}\Omega_{ij,h}^{-1}\Sigma_{ij\cdot-i}\bar{\beta}$.

For consistency of the Granger-causality measure, suppose in addition that

$$k_{\Theta}^3 k_u^3 k_A^5 Q_n = o(1). \quad (4.15)$$

This condition controls estimation of the partialled-out source covariance and the full-projection residual variance entering $s_{ij}(h)$.

THEOREM 4.2 (Consistency of the Granger-causality measure). *Under Assumption 4.1, (4.11), and (4.15),*

$$\hat{s}_{ij}(h) \xrightarrow{P} s_{ij}(h). \quad (4.16)$$

The result follows from joint consistency of $\hat{\beta}_{ij,h}$, $\hat{\Sigma}_{ij\cdot-i}$, and $\hat{\sigma}_f^2(h)$. Proofs are in Appendix A.2.

5. MONTE CARLO SIMULATIONS

We examine the finite-sample behavior of $\hat{\beta}_{ij,h}$ and $\hat{s}_{ij}(h)$ under the two grouping schemes of Section 3.2. We consider $T \in \{500, 1000, 1500\}$, horizons $h \in \{1, 5, 10\}$, and $R = 2000$ Monte Carlo replications. Population targets are computed from the stationary covariance implied by the DGP. For $\beta_{ij,h}$, we report the Euclidean norm of the bias, $\|\mathbb{E}[\hat{\beta}_{ij,h}] - \beta_{ij,h}\|_2$, the root mean squared Euclidean error $\{\mathbb{E}\|\hat{\beta}_{ij,h} - \beta_{ij,h}\|_2^2\}^{1/2}$, and componentwise 95% coverage probabilities. For $s_{ij}(h)$, we report its population value, bias, and RMSE.

5.1. Simulation designs

Design A: HAR grouping. The first design uses the HAR transformation in (3.5). We set $d = 100$ and $p = 22$, so that $X_t = GY_t \in \mathbb{R}^{300}$. Writing the one-step grouped coefficient matrix as $B =$

$[B^{(d)}, B^{(w)}, B^{(m)}]$, with each block in $\mathbb{R}^{100 \times 100}$, we set

$$B_{ij}^{(d)} = 0.12 \mathbf{1}\{|i - j| \leq 1\}, \quad B_{ij}^{(w)} = 0.08 \mathbf{1}\{|i - j| \leq 1\}, \quad B_{ij}^{(m)} = 0.05 \mathbf{1}\{|i - j| \leq 1\}. \quad (5.1)$$

The unrestricted VAR(22) coefficients are obtained from $[A_1, \dots, A_{22}] = BG$. Innovations are Gaussian with $\Sigma_{u,ij} = 0.2^{|i-j|}$. The design therefore combines the daily, weekly, and monthly aggregation of the HAR state with cross-sectional dependence that decays with the distance between variables.

Design B: grouped-index VAR. The second design uses receiver-group-specific predictive states. We set $\bar{g} = 3$, $n_g = 50$, and $p = 3$, so that $d = 150$. For each external group $k \neq g$, we use the equal-weight index $y_{gk,t}^* = 50^{-1} \sum_{i \in \mathcal{G}_k} y_{i,t}$. For $j \in \mathcal{G}_g$,

$$y_{j,t} = \sum_{\ell=1}^3 \sum_{i \in \mathcal{G}_g} a_{ji,\ell} y_{i,t-\ell} + \sum_{\ell=1}^3 \sum_{\substack{k=1 \\ k \neq g}}^3 b_{jk,\ell} y_{gk,t-\ell}^* + u_{j,t}. \quad (5.2)$$

At lags $\ell = 1, 2, 3$, the diagonal within-group coefficients are 0.20, 0.14, and 0.08, while the corresponding first off-diagonal coefficients are 0.10, 0.07, and 0.04. All remaining within-group coefficients are zero. The external-group coefficients satisfy $b_{jk,\ell} = 0.01$ at each lag. Since each group index averages 50 variables, the corresponding entries of an unrestricted cross-group block are 0.0002. Innovations are Gaussian with $\Sigma_{u,ij} = 0.2^{|i-j|}$.

Receivers in the same group share $G_j = G_{g(j)}$. At each lag, the predictive state contains the 50 variables in the receiver's own group and one index for each of the two external groups, giving $r_j = 3(50 + 2) = 156$. We report results for the directed link $i = 2 \rightarrow j = 1$, with $i, j \in \mathcal{G}_1$, so that $\beta_{21,h} \in \mathbb{R}^3$.

For both designs, the one-step equations are estimated equation by equation using ℓ_1 penalization with tuning parameters selected by BIC; Design B uses the adaptive version. The estimated coefficient and innovation covariance matrices are thresholded as described above and used to construct the required nuisance quantities. We estimate $s_{ij}(h)$ using (4.5), based on the full-projection

residual variance and the estimated partial covariance $(R_{ij} \hat{\Theta}_j R'_{ij})^{-1}$; no additional restricted regression is therefore required. Newey–West standard errors are used for inference.

5.2. Results

Tables 1–4 summarize the finite-sample performance of the proposed estimators under the HAR and grouped-index VAR designs, including configurations with positive population strength and configurations for which population strength is reported as zero. Across all four tables, the RMSE of both the coefficient estimator and the strength estimator decreases monotonically with the sample size at every horizon. The coefficient RMSE at $T = 1500$ is approximately 51%–59% of its value at $T = 500$, broadly consistent with the improvement associated with a $T^{-1/2}$ rate. The norm of the coefficient bias is generally small relative to the corresponding RMSE, indicating that sampling variation accounts for most of the reported mean squared error. However, coefficient bias does not decline monotonically in every configuration, particularly under the HAR grouping; the evidence for improved precision is therefore stronger than the evidence for a uniform reduction in bias.

The coefficient results also reveal differences in the sensitivity of the two designs to the forecast horizon. In the positive-strength HAR configuration, RMSE_2 decreases from 0.2512 to 0.1371 at $h = 1$ as T increases from 500 to 1500, whereas the corresponding decline at $h = 10$ is from 0.4068 to 0.2353. A similar increase in uncertainty with the horizon appears in the zero-strength HAR configuration. In the grouped-index VAR design, coefficient RMSE is smaller and varies less across horizons: at $T = 1500$, it ranges from 0.0464 to 0.0514 in the positive-strength configuration and from 0.0473 to 0.0507 in the zero-strength configuration. The more pronounced horizon dependence under the HAR grouping is consistent with the greater difficulty of estimating multi-horizon projections on temporally aggregated predictors, although the tables do not isolate the contribution of regressor dependence from other features of the design. In particular, differences in predictive-state dimension, coefficient scaling, dynamics, and first-step regularization preclude interpreting the cross-design RMSE comparison as a ranking of the two grouping schemes.

Componentwise coverage is generally close to its nominal level in the grouped-index VAR design, ranging from 0.937 to 0.9585 across the reported configurations. The HAR results are less uniform. Coverage for several components is close to 95%, but the third component in the zero-strength configuration exhibits appreciable undercoverage at longer horizons. At $h = 5$, its coverage rises from 0.909 at $T = 500$ to 0.9305 at $T = 1500$; at $h = 10$, it remains 0.928 at the largest sample size. With 2,000 replications, the Monte Carlo standard error of a coverage estimate near 0.95 is approximately 0.0049, so these shortfalls cannot reasonably be attributed solely to simulation noise. They indicate that the feasible normal approximation with HAC standard errors is less accurate for these components. The results thus provide stronger evidence of accurate marginal inference for the grouped-index VAR design than for the longer-horizon HAR specifications.

The population strength values reported in Tables 1 and 3 illustrate the importance of distinguishing the forecast horizon from the presence of a predictive link. In the HAR design, population strength declines from 0.0438 at $h = 1$ to 0.0028 at $h = 5$ and 0.0004 at $h = 10$. The corresponding values in the grouped-index VAR design are 0.0212, 0.0072, and 0.0011. Thus, for the links considered, the source's incremental contribution relative to the full-projection forecast-error variance diminishes substantially as the horizon increases. This attenuation is consistent with the propagation and eventual decay of predictive effects under the specified stable dynamics. It is a feature of these simulation designs, however, rather than a general implication that horizon-specific Granger-causality strength must decrease monotonically.

The strength estimator exhibits positive bias in every reported configuration, with both bias and RMSE declining as the sample size increases. Its relative accuracy nevertheless depends strongly on the magnitude of the population signal. For the positive-strength HAR case at $h = 1$, bias falls from 0.0015 to 0.0011 and RMSE from 0.0180 to 0.0107 between $T = 500$ and $T = 1500$. At $h = 10$, the bias at $T = 1500$ is 0.0031, compared with a population strength of only 0.0004. Similarly, the grouped-index VAR bias at $h = 10$ remains 0.0023 when population strength is 0.0011. Consequently, a smaller absolute RMSE at a longer horizon need not imply more accu-

rate measurement relative to the underlying signal. The observed upward bias is consistent with the quadratic-form construction in (4.5): squaring estimated coefficients introduces a contribution from estimation noise, even when their first-order bias is small. This provides a plausible explanation for the prominence of bias near zero, without attributing the entire finite-sample bias to a single source.

The configurations with zero reported population strength in Tables 2 and 4 make this distinction particularly clear. At an exact null, the quadratic-form estimator is nonnegative, and sampling variation in $\hat{\beta}_{ij,h}$ can therefore generate positive estimated strength without a population predictive link. In the HAR configuration at $h = 1$, strength bias decreases from 0.0058 to 0.0029 and then 0.0019 as the sample size increases; the corresponding grouped-index VAR values are 0.0059, 0.0030, and 0.0020. These declines are close to an inverse-sample-size pattern. Under a fixed-dimensional root- T approximation for the source coefficient block, this pattern is consistent with strength being locally quadratic at the null, so that its leading contribution from coefficient estimation error is of order T^{-1} . This interpretation also explains why approximately unbiased coefficient estimates need not produce approximately unbiased strength estimates in small samples.

Taken together, the simulations document improving estimation accuracy as the sample size increases, while identifying weak-signal strength estimation and longer-horizon HAR coverage as the principal finite-sample limitations in the reported experiments. They also motivate the distinction between measuring predictive strength and determining whether a link is statistically supported: a positive value of $\hat{s}_{ij}(h)$ alone is insufficient evidence of an active edge. The use of coefficient-based testing to screen estimated strengths is consistent with this distinction. Nevertheless, the tables report marginal confidence-interval coverage rather than joint Wald rejection frequencies or network-level error rates. They therefore do not directly establish the size or power of the joint link tests, the effectiveness of multiple-testing adjustments, or the finite-sample accuracy of the screened network weights.

Table 1: Monte Carlo results: HAR grouping.

h		$T = 500$	$T = 1000$	$T = 1500$
1	$\ \text{Bias}\ _2$	0.0275	0.0177	0.0146
	RMSE_2	0.2512	0.1692	0.1371
	95% coverage	(0.934, 0.946, 0.952)	(0.9375, 0.9495, 0.9585)	(0.938, 0.9475, 0.958)
	True $s_{ij}(h)$	0.0438	0.0438	0.0438
	$s_{ij}(h)$ bias	0.0015	0.0014	0.0011
	$s_{ij}(h)$ RMSE	0.018	0.0133	0.0107
5	$\ \text{Bias}\ _2$	0.0139	0.0191	0.0177
	RMSE_2	0.3788	0.2622	0.2153
	95% coverage	(0.946, 0.929, 0.934)	(0.9445, 0.945, 0.942)	(0.9445, 0.953, 0.947)
	True $s_{ij}(h)$	0.0028	0.0028	0.0028
	$s_{ij}(h)$ bias	0.0075	0.0042	0.003
	$s_{ij}(h)$ RMSE	0.0109	0.0063	0.0048
10	$\ \text{Bias}\ _2$	0.005	0.0134	0.0125
	RMSE_2	0.4068	0.2844	0.2353
	95% coverage	(0.9465, 0.935, 0.924)	(0.942, 0.942, 0.9315)	(0.9505, 0.9375, 0.933)
	True $s_{ij}(h)$	0.0004	0.0004	0.0004
	$s_{ij}(h)$ bias	0.0082	0.0043	0.0031
	$s_{ij}(h)$ RMSE	0.0112	0.0059	0.0042

Notes: Results are based on 2,000 replications. $\|\text{Bias}\|_2$ is the Euclidean norm of the componentwise bias vector, and RMSE_2 is $\{\mathbb{E}\|\hat{\beta}_{ij,h} - \beta_{ij,h}\|_2^2\}^{1/2}$. Coverage probabilities are reported separately for the three components of $\beta_{ij,h}$.

6. EMPIRICAL APPLICATIONS

We illustrate the horizon-structure network in two applications to large U.S. equities. The first studies daily volatility transmission across sectors using the HAR grouping. The second studies monthly equity returns using the grouped-index VAR and asks whether systematic predictive connectedness contains information about subsequent equity-market tail risk. The two applications therefore emphasize different dimensions of the framework: the first identifies where predictive transmission originates and where it is received, while the second summarizes cross-sector transmission into a system-wide measure and evaluates its predictive content.

We use historical constituents of the S&P 100 whenever point-in-time membership is available. Sector assignments are based on a time-consistent industry classification. If historical GICS

Table 2: Monte Carlo results: HAR grouping.

h		$T = 500$	$T = 1000$	$T = 1500$
1	$\ \text{Bias}\ _2$	0.0029	0.004	0.0037
	RMSE_2	0.2747	0.1789	0.1411
	95% coverage	(0.9505, 0.9445, 0.9265)	(0.944, 0.9405, 0.939)	(0.942, 0.9525, 0.9395)
	True $s_{ij}(h)$	0.	0.	0.
	$s_{ij}(h)$ bias	0.0058	0.0029	0.0019
	$s_{ij}(h)$ RMSE	0.0076	0.0037	0.0024
5	$\ \text{Bias}\ _2$	0.0073	0.0073	0.0084
	RMSE_2	0.4	0.2764	0.2215
	95% coverage	(0.9435, 0.95, 0.909)	(0.949, 0.9485, 0.918)	(0.9565, 0.946, 0.9305)
	True $s_{ij}(h)$	0.00	0.00	0.00
	$s_{ij}(h)$ bias	0.0083	0.0043	0.0029
	$s_{ij}(h)$ RMSE	0.011	0.0057	0.0039
10	$\ \text{Bias}\ _2$	0.0118	0.0102	0.0073
	RMSE_2	0.4152	0.296	0.239
	95% coverage	(0.945, 0.9475, 0.9155)	(0.944, 0.942, 0.91)	(0.9515, 0.941, 0.928)
	True $s_{ij}(h)$	0.00	0.00	0.00
	$s_{ij}(h)$ bias	0.0088	0.0048	0.0032
	$s_{ij}(h)$ RMSE	0.0118	0.0063	0.0043

Notes: Results are based on 2,000 replications. $\|\text{Bias}\|_2$ is the Euclidean norm of the componentwise bias vector, and RMSE_2 is $\{\mathbb{E}\|\hat{\beta}_{ij,h} - \beta_{ij,h}\|_2^2\}^{1/2}$. Coverage probabilities are reported separately for the three components of $\beta_{ij,h}$.

classifications are used, changes in the GICS taxonomy are harmonized so that sector definitions remain comparable across rolling windows. Firms without sufficient return histories for a given estimation window are excluded from that window.

6.1. Sectoral transmission of daily volatility

The first application studies whether volatility originating in one sector contains incremental predictive information for subsequent volatility in other sectors, and whether the direction of this transmission changes with the forecast horizon.

Data and volatility measure. We collect daily open, high, low, and close prices for S&P 100 constituents from Bloomberg or Refinitiv and match firms to their sector classifications. Let O_{it} , H_{it} , L_{it} , and C_{it} denote the open, high, low, and close prices of firm i on trading day t . Our baseline

Table 3: Monte Carlo results: grouped-index VAR.

h		$T = 500$	$T = 1000$	$T = 1500$
1	$\ \text{Bias}\ _2$	0.0019	0.0004	0.0005
	RMSE_2	0.0829	0.058	0.0464
	95% coverage	(0.9445, 0.937, 0.946)	(0.9395, 0.9475, 0.947)	(0.9445, 0.9475, 0.9525)
	True $s_{ij}(h)$	0.0212	0.0212	0.0212
	$s_{ij}(h)$ bias	0.0081	0.0041	0.0026
	$s_{ij}(h)$ RMSE	0.0164	0.0106	0.0083
5	$\ \text{Bias}\ _2$	0.0056	0.0036	0.0019
	RMSE_2	0.0867	0.0603	0.0492
	95% coverage	(0.9535, 0.9545, 0.954)	(0.9515, 0.9495, 0.954)	(0.956, 0.948, 0.9565)
	True $s_{ij}(h)$	0.0072	0.0072	0.0072
	$s_{ij}(h)$ bias	0.0074	0.0036	0.0024
	$s_{ij}(h)$ RMSE	0.0136	0.0082	0.0063
10	$\ \text{Bias}\ _2$	0.0033	0.0027	0.0014
	RMSE_2	0.088	0.0625	0.0514
	95% coverage	(0.9565, 0.953, 0.953)	(0.944, 0.952, 0.958)	(0.9415, 0.948, 0.954)
	True $s_{ij}(h)$	0.0011	0.0011	0.0011
	$s_{ij}(h)$ bias	0.0066	0.0035	0.0023
	$s_{ij}(h)$ RMSE	0.0097	0.0055	0.0039

Notes: Results are based on 2,000 replications. $\|\text{Bias}\|_2$ is the Euclidean norm of the componentwise bias vector, and RMSE_2 is $\{\mathbb{E}\|\hat{\beta}_{21,h} - \beta_{21,h}\|_2^2\}^{1/2}$. Coverage probabilities are reported separately for the three components of $\beta_{21,h}$.

daily variance proxy is the Garman–Klass estimator ([Garman and Klass, 1980](#)),

$$v_{it} = \frac{1}{2} \left[\log \left(\frac{H_{it}}{L_{it}} \right) \right]^2 - (2 \log 2 - 1) \left[\log \left(\frac{C_{it}}{O_{it}} \right) \right]^2. \quad (6.1)$$

We set $y_{it} = \log v_{it}$ and center each series within the estimation window. Observations for which v_{it} cannot be constructed are treated as missing rather than set to zero. As a return-only robustness measure, replace v_{it} by the squared close-to-close return r_{it}^2 , which provides a noisier proxy for daily return variance.

HAR network. For each rolling estimation window, construct the HAR predictive state of Section 3.2.1. Thus the source block for firm i contains its daily volatility and its 5- and 22-trading-day averages, as in [Corsi \(2009\)](#). The baseline estimation window contains 1,000 trading days. We

Table 4: Monte Carlo results: grouped-index VAR.

h		$T = 500$	$T = 1000$	$T = 1500$
1	$\ \text{Bias}\ _2$	0.0016	0.0008	0.0008
	RMSE_2	0.0816	0.058	0.0473
	95% coverage	(0.954, 0.948, 0.943)	(0.9535, 0.954, 0.941)	(0.9440, 0.9485, 0.9405)
	True $s_{ij}(h)$	0.	0.	0.
	$s_{ij}(h)$ bias	0.0059	0.003	0.002
	$s_{ij}(h)$ RMSE	0.0075	0.0039	0.0026
5	$\ \text{Bias}\ _2$	0.0021	0.0013	0.0006
	RMSE_2	0.0867	0.0618	0.0507
	95% coverage	(0.955, 0.9455, 0.9495)	(0.947, 0.947, 0.951)	(0.943, 0.952, 0.9515)
	True $s_{ij}(h)$	0.00	0.00	0.00
	$s_{ij}(h)$ bias	0.0063	0.0033	0.0022
	$s_{ij}(h)$ RMSE	0.0083	0.0043	0.003
10	$\ \text{Bias}\ _2$	0.0013	0.0011	0.0012
	RMSE_2	0.0897	0.0623	0.0506
	95% coverage	(0.9585, 0.9555, 0.95)	(0.9535, 0.953, 0.9555)	(0.9505, 0.9535, 0.9545)
	True $s_{ij}(h)$	0.00	0.00	0.00
	$s_{ij}(h)$ bias	0.0066	0.0033	0.0022
	$s_{ij}(h)$ RMSE	0.0086	0.0043	0.0029

Notes: Results are based on 2,000 replications. $\|\text{Bias}\|_2$ is the Euclidean norm of the componentwise bias vector, and RMSE_2 is $\{\mathbb{E}\|\hat{\beta}_{ij,h} - \beta_{ij,h}\|_2^2\}^{1/2}$. Coverage probabilities are reported separately for the three components of $\beta_{ij,h}$.

report results for $h \in \{1, 5, 22\}$ trading days.

The horizons refer to the date of the forecast target. In particular, $h = 5$ measures predictive content for daily volatility five trading days ahead; it should not be interpreted as a five-day aggregate volatility forecast. The daily, weekly, and monthly terminology in the HAR specification refers instead to the temporal aggregation of the predictors.

For each ordered firm pair (i, j) , estimate $\hat{s}_{ij}(h)$ and test $H_0(i \rightarrow j, h)$ using the procedures of Sections 3 and 4. The estimated binary edge is $\hat{c}_{ij}(h)$. Because the binary network involves many simultaneous firm-level tests, the baseline binary results control the false discovery rate within each window and horizon; unadjusted 5% tests are reported as a robustness exercise. The weighted measures do not require thresholding.

Sector-to-sector transmission. For two distinct sectors $\mathcal{G}_a$ and $\mathcal{G}_b$, calculate $\widehat{\text{WGC}}_{\mathcal{G}_a \rightarrow \mathcal{G}_b}(h)$ ac-

according to (2.4). The principal empirical object is the sector-to-sector WGC matrix. Rows denote source sectors and columns receiver sectors, so that the (a, b) entry measures average incremental predictive transmission from firms in $\mathcal{G}_a$ to firms in $\mathcal{G}_b$.

We report separate heat maps for $h = 1, 5$, and 22 . If within-sector transmission is displayed on the diagonal, it is normalized by $N_a(N_a - 1)$ and excludes self-links. The off-diagonal elements are the main objects of interest.

For each sector, we also calculate $\widehat{\text{WGC}}_{\mathcal{G}_a}^{\text{out}}(h)$ and $\widehat{\text{WGC}}_{\mathcal{G}_a}^{\text{in}}(h)$, together with their difference. The corresponding binary GC measures are reported as a complement. These statistics reveal whether a sector is predominantly a transmitter or a receiver of volatility and whether its position changes with the horizon.

Finally, rolling estimation produces a time series of sectoral transmission. We plot the main sector-to-sector links and the outward and inward measures around episodes of market stress. Of particular interest is whether increases in cross-sector volatility transmission first appear at short horizons and subsequently extend to longer horizons.

Empirical results. The sector-to-sector profiles are constructed from a fixed panel of 77 firms assigned to nine sectors, using 2,283 rolling windows of 252 trading-day observations. The window-end dates extend from 1 May 2017 to 29 May 2026. For the results reported here, a directed firm-level link is retained when its unadjusted Wald-test p -value is below 0.05, and its weight is $\hat{c}_{ij}(h)\hat{s}_{ij}(h)$. Each sector-pair measure averages these weights over all possible firm pairs between the two sectors, including pairs with zero estimated weight. For a focal sector $\mathcal{G}_a$ and counterpart $\mathcal{G}_b$, the Out panel reports $\widehat{\text{WGC}}_{\mathcal{G}_a \rightarrow \mathcal{G}_b}(h)$, the In panel reports the reverse direction, and Net is their difference. Each counterpart is displayed separately. Positive Net therefore indicates stronger estimated transmission from the focal sector to that particular counterpart, while negative Net indicates the reverse asymmetry. Representative bilateral profiles are presented in Figures 1–4.

The first feature of the results is a pronounced horizon dependence in the level of estimated weighted transmission. Averaging equally across the 72 ordered sector pairs and all rolling win-

dows gives mean WGC values of 0.0195, 0.0236, and 0.0315 at $h = 1, 5$, and 22, respectively. These figures are descriptive averages of the bilateral measures, rather than sums of sectoral transmission. The corresponding medians across sector-pair–window observations are 0.0037, 0.0047, and 0.0065. Moreover, the time-average WGC at $h = 22$ exceeds its one-day counterpart for every ordered sector pair. Thus, the increase in average estimated transmission at longer horizons is broadly distributed across the network. It is consistent with volatility histories containing information about subsequent daily volatility that is not fully captured by their one-day predictive contribution. Because the target remains daily volatility at $t + h$, this pattern does not represent an accumulation of volatility over the intervening h days. Nor does it imply that an individual pair’s transmission increases with the horizon in every rolling window.

The sectoral composition of transmission is heterogeneous. At the one-day horizon, Energy has a positive time-average Net measure against each of the other eight sectors. For example, average transmission from Energy to Information Technology is 0.0265, compared with 0.0193 in the reverse direction. Energy retains positive average Net transmission against all eight counterparts at $h = 22$, while Utilities has positive average Net transmission against seven counterparts at that horizon. Among the larger 22-day directional means are Utilities to Communication Services (0.0399, compared with 0.0289 in reverse) and Energy to Consumer Discretionary (0.0388, compared with 0.0292 in reverse). Communication Services, by contrast, has negative time-average Net transmission against every counterpart at both $h = 1$ and $h = 22$. These findings identify asymmetries in average conditional predictive content; they do not establish that the same direction dominates continuously throughout the sample. Figure 1 displays the Energy–Information Technology profiles across the three horizons.

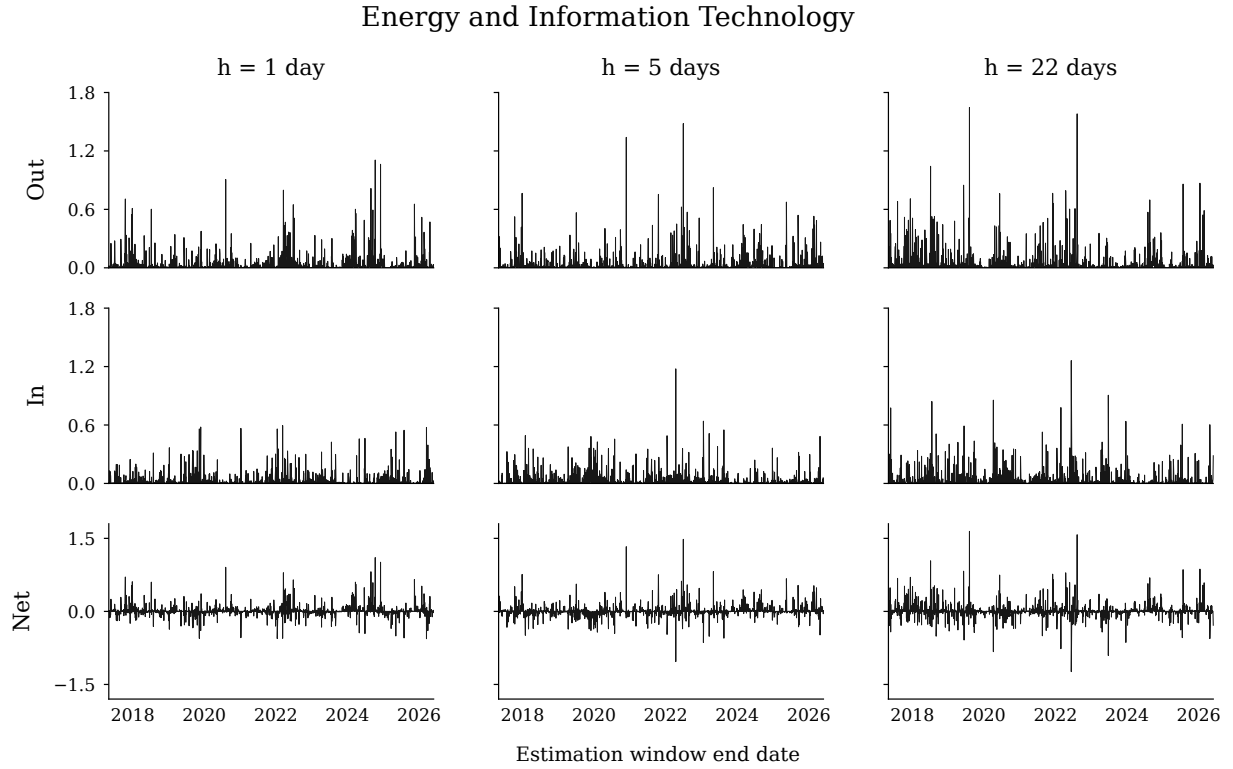


Figure 1: Rolling bilateral transmission: Energy and Information Technology. Energy is the focal sector: Out denotes Energy to Information Technology, In denotes the reverse direction, and Net equals Out minus In. Columns show forecasts of daily log variance 1, 5, and 22 trading days ahead. Weights are unadjusted 5% Wald links multiplied by estimated strengths and averaged over all 27 possible directed firm pairs. The 2,283 estimates use rolling 252-observation windows. Values are in natural-log strength units. Out and In share a scale across horizons; Net uses the same magnitude symmetrically. Scales differ across figures.

The forecast horizon also changes the direction of some bilateral asymmetries. For Utilities relative to Financials, the average Net measure is slightly negative at $h = 1$ (-0.0007), but becomes positive at $h = 5$ (0.0036) and $h = 22$ (0.0080). Similarly, Health Care has negative average Net transmission relative to Information Technology at one day (-0.0015), but positive values at five and 22 days (0.0069 and 0.0040). These reversals show that a sector's role as a transmitter or receiver depends on both its counterpart and the forecast horizon. They are compatible with differences in the persistence and indirect propagation of predictive information across sectors, although the estimated networks do not identify the underlying economic mechanisms. The signs of these sample averages are descriptive; no separate test of the significance of their differences is reported

here. These horizon-dependent changes are illustrated in Figures 2 and 3.

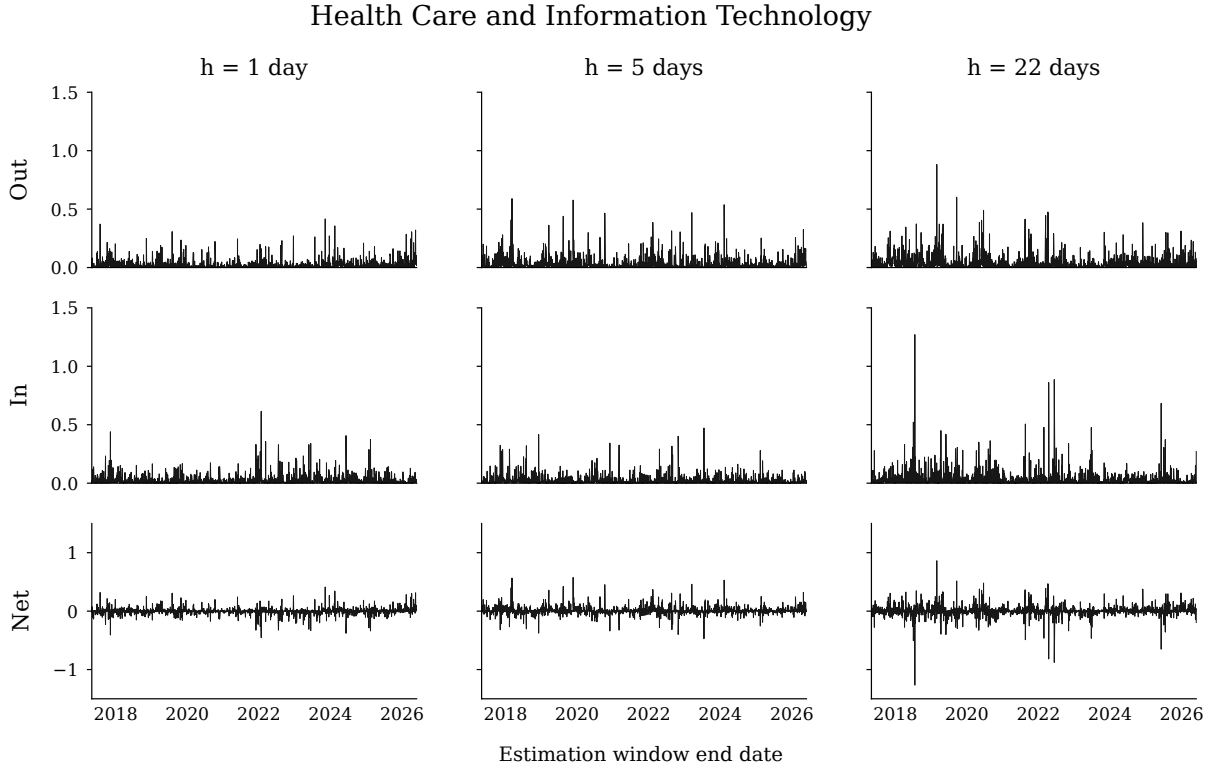


Figure 2: Rolling bilateral transmission: Health Care and Information Technology. Health Care is the focal sector: Out denotes Health Care to Information Technology, In denotes the reverse direction, and Net equals Out minus In. Columns show forecasts of daily log variance 1, 5, and 22 trading days ahead. Weights are unadjusted 5% Wald links multiplied by estimated strengths and averaged over all 117 possible directed firm pairs. The 2,283 estimates use rolling 252-observation windows. Values are in natural-log strength units. Out and In share a scale across horizons; Net uses the same magnitude symmetrically. Scales differ across figures.

The rolling profiles further show that average transmission is affected by intermittent large observations. At each horizon, the mean WGC across sector-pair-window observations is approximately five times its median, indicating a markedly right-skewed distribution. The distinction between the magnitude and frequency of directional dominance is particularly clear for Utilities relative to Financials at $h = 22$. Although the time-average Net measure is positive at approximately 0.0080, Utilities is a net transmitter to Financials in only 37.1% of windows and a net receiver in 62.3%, with the remainder recording zero Net transmission. Its positive average therefore reflects larger positive episodes rather than more frequent positive observations. Accordingly,

a classification based solely on average Net transmission can obscure substantial changes in bilateral roles over time. Figure 3 shows the bilateral profiles underlying this distinction.

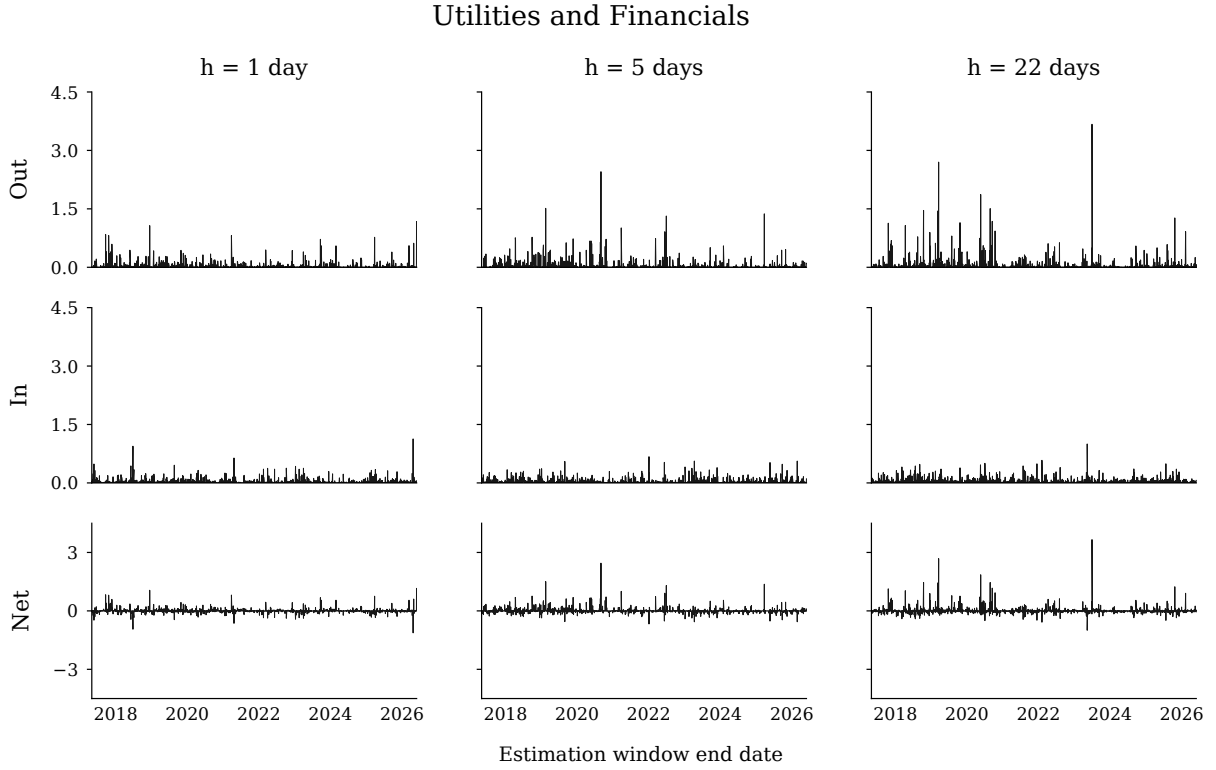


Figure 3: Rolling bilateral transmission: Utilities and Financials. Utilities is the focal sector: Out denotes Utilities to Financials, In denotes the reverse direction, and Net equals Out minus In. Columns show forecasts of daily log variance 1, 5, and 22 trading days ahead. Weights are unadjusted 5% Wald links multiplied by estimated strengths and averaged over all 42 possible directed firm pairs. The 2,283 estimates use rolling 252-observation windows. Values are in natural-log strength units. Out and In share a scale across horizons; Net uses the same magnitude symmetrically. Scales differ across figures.

The timing of these episodes also differs across sector pairs and horizons. For example, one-day transmission from Energy to Financials averages 0.0701 over windows ending in 2024, compared with 0.0162 over windows ending in 2023. At the 22-day horizon, the window ending 30 June 2023 produces a pronounced outward-transmission spike from Utilities to several counterparts, including Energy and Industrials. These observations illustrate that elevated transmission can be concentrated in particular directions and estimation windows rather than constituting a uniform increase throughout the network. Their timing must nevertheless be interpreted in light of the

rolling design: each point summarizes a fitted relationship over a 252-observation window, and adjacent windows overlap extensively. A spike dated at a window endpoint therefore does not identify a structural shock occurring on that date, and the figures alone do not establish a sequence in which transmission moves from short to long horizons. Figure 4 displays the Energy–Financials profiles discussed above.

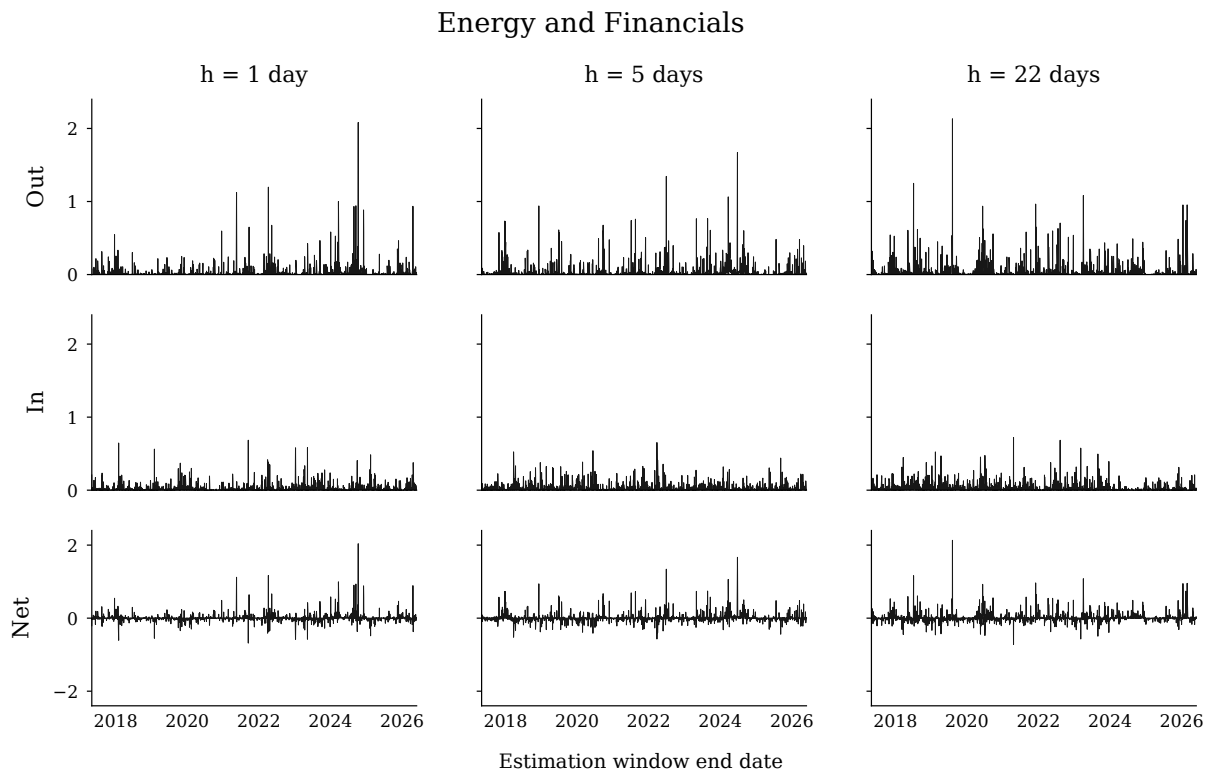


Figure 4: Rolling bilateral transmission: Energy and Financials. Energy is the focal sector: Out denotes Energy to Financials, In denotes the reverse direction, and Net equals Out minus In. Columns show forecasts of daily log variance 1, 5, and 22 trading days ahead. Weights are unadjusted 5% Wald links multiplied by estimated strengths and averaged over all 42 possible directed firm pairs. The 2,283 estimates use rolling 252-observation windows. Values are in natural-log strength units. Out and In share a scale across horizons; Net uses the same magnitude symmetrically. Scales differ across figures.

The interpretation of these patterns is conditional on the measurement and inference choices. Normalization by the number of possible firm pairs makes WGC a measure of average screened predictive strength, rather than a sector’s total economic contribution to market-wide risk. This distinction is especially relevant for Energy and Utilities, each of which contains only three firms

in the panel: bilateral averages involving these sectors may be sensitive to individual firm links. In addition, the unadjusted 5% tests used here do not control the false discovery rate across the network, and the quadratic-form strength estimator can be upward biased when predictive signals are weak, as illustrated by the Monte Carlo results. The empirical windows are also shorter than the smallest sample size considered in those simulations. Consequently, the higher long-horizon WGC values and prominent isolated spikes should be interpreted as features of the estimated predictive network, with their economic interpretation conditional on robustness to multiple-testing adjustment, window length, and influential observations.

7. CONCLUSION

This paper develops a framework for measuring and conducting inference on the horizon structure of financial contagion in high-dimensional predictive systems. Contagion strength is defined through the incremental reduction in forecast-error variance attributable to a source firm’s information, conditional on the remaining predictors. The resulting scale-free measure distinguishes the magnitude of a predictive contribution from the statistical evidence supporting its presence. Combining estimated strengths with indicators of statistically detected links yields directed, weighted networks whose firm-level and sector-level transmission patterns can be compared across forecast horizons. This formulation accommodates grouped predictive specifications, including the HAR and grouped-index VAR models, and makes the forecast horizon an explicit dimension of network analysis.

The methodological analysis addresses the fact that sparsity in underlying one-step dynamics need not be preserved under predictor aggregation or multi-step propagation. By formulating the estimation problem directly in the grouped predictive system, the proposed approach provides conditions for asymptotic normality of the debiased coefficient estimator, feasible Wald inference for individual source–receiver relationships, and consistency of the estimated transmission strength. Accounting for serial dependence is central to inference at multiple horizons. The analysis also

distinguishes coefficient inference from strength estimation: because the strength functional is locally quadratic at the null, small coefficient estimation errors can generate positive estimated strengths even in the absence of population transmission.

ACKNOWLEDGMENTS

This work was supported by the research funds provided by the University of Mannheim.

REFERENCES

- ABRAMOVICH, F., Y. BENJAMINI, D. L. DONOHO, AND I. M. JOHNSTONE (2006): “Adapting to unknown sparsity by controlling the false discovery rate,” *The Annals of Statistics*, 34, 584–653.
- BARIGOZZI, M. AND C. BROWNLEES (2019): “Nets: Network estimation for time series,” *Journal of Applied Econometrics*, 34, 347–364.
- BARUNÍK, J. AND T. KŘEHLÍK (2018): “Measuring the frequency dynamics of financial connectedness and systemic risk,” *Journal of Financial Econometrics*, 16, 271–296.
- BENJAMINI, Y. AND Y. HOCHBERG (1995): “Controlling the false discovery rate: a practical and powerful approach to multiple testing,” *Journal of the Royal statistical society: series B (Methodological)*, 57, 289–300.
- BICKEL, P. J. AND E. LEVINA (2008): “Covariance regularization by thresholding,” *The Annals of Statistics*, 36, 2577 – 2604.
- BILLIO, M., M. GETMANSKY, A. W. LO, AND L. PELIZZON (2012): “Econometric measures of connectedness and systemic risk in the finance and insurance sectors,” *Journal of Financial Economics*, 104, 535–559.

- CAI, T. AND W. LIU (2011): “Adaptive thresholding for sparse covariance matrix estimation,” *Journal of the American Statistical Association*, 106, 672–684.
- CHEN, J., D. LI, Y.-N. LI, AND O. LINTON (2025): “Estimating time-varying networks for high-dimensional time series,” *Journal of Econometrics*, 249, 105941.
- CHERNOZHUKOV, V., D. CHETVERIKOV, M. DEMIRER, E. DUFLO, C. HANSEN, W. NEWEY, AND J. ROBINS (2018): “Double/debiased machine learning for treatment and structural parameters,” *Econometrics Journal*, 21, C1–C68.
- CORSI, F. (2009): “A simple approximate long-memory model of realized volatility,” *Journal of Financial Econometrics*, 7, 174–196.
- CORSI, F., F. LILLO, D. PIRINO, AND L. TRAPIN (2018): “Measuring the propagation of financial distress with granger-causality tail risk networks,” *Journal of Financial Stability*, 38, 18–36.
- DEMIRER, M., F. X. DIEBOLD, L. LIU, AND K. YILMAZ (2018): “Estimating global bank network connectedness,” *Journal of Applied Econometrics*, 33, 1–15.
- DETTAA, E. AND E. WANG (2026): “Sparse VARs Do Not Imply Sparse Local Projections: Robust Inference for High-Dimensional Granger Causality,” *arXiv preprint arXiv:2410.04330*.
- DIEBOLD, F. X. AND K. YILMAZ (2014): “On the network topology of variance decompositions: Measuring the connectedness of financial firms,” *Journal of Econometrics*, 182, 119–134.
- DUFOUR, J.-M. AND E. RENAULT (1998): “Short run and long run causality in time series: theory,” *Econometrica*, 66, 1099–1125.
- DUFOUR, J.-M. AND A. TAAMOUTI (2010): “Short and long run causality measures: Theory and inference,” *Journal of Econometrics*, 154, 42–58.

- FORBES, K. J. AND R. RIGOBON (2002): “No contagion, only interdependence: measuring stock market comovements,” *The Journal of Finance*, 57, 2223–2261.
- FRANCH, F., L. NOCCIOLA, AND A. VOULDIS (2024): “Temporal networks and financial contagion,” *Journal of Financial Stability*, 71, 101224.
- GARMAN, M. B. AND M. J. KLASS (1980): “On the Estimation of Security Price Volatilities from Historical Data,” *Journal of Business*, 53, 67–78.
- GEWEKE, J. (1982): “Measurement of linear dependence and feedback between multiple time series,” *Journal of the American Statistical Association*, 77, 304–313.
- (1984): “Measures of conditional linear dependence and feedback between time series,” *Journal of the American Statistical Association*, 79, 907–915.
- HECQ, A., L. MARGARITELLA, AND S. SMEEKES (2023): “Granger causality testing in high-dimensional VARs: a post-double-selection procedure,” *Journal of Financial Econometrics*, 21, 915–958.
- KRAMPE, J., E. PAPARODITIS, AND C. TRENKLER (2023): “Structural inference in sparse high-dimensional vector autoregressions,” *Journal of Econometrics*, 234, 276–300.
- PAPANA, A., C. KYRTSOU, D. KUGIUMTZIS, AND C. DIKS (2017): “Financial networks based on Granger causality: A case study,” *Physica A*, 482, 65–73.
- PESARAN, M. H. AND A. PICK (2007): “Econometric issues in the analysis of contagion,” *Journal of Economic Dynamics and Control*, 31, 1245–1277.
- UEMATSU, Y. AND T. YAMAGATA (2025): “Discovering the network Granger causality in large vector autoregressive models,” *Journal of the American Statistical Association*, 120, 2385–2396.
- ZHU, X. AND R. PAN (2020): “Grouped Network Vector Autoregression,” *Statistica Sinica*, 30, 1437–1462.

ZHU, X., R. PAN, G. LI, Y. LIU, AND H. WANG (2017): “Network vector autoregression,” *The Annals of Statistics*, 45, 1096–1123.

ZOU, H. (2006): “The adaptive lasso and its oracle properties,” *Journal of the American statistical association*, 101, 1418–1429.

ZOU, H., T. HASTIE, AND R. TIBSHIRANI (2007): “On the “degrees of freedom” of the lasso,” *The Annals of Statistics*, 35, 2173 – 2192.

Online Appendix

to “The Horizon Structure of Contagion: High-Dimensional Measurement and Inference”

Abderrahim Taamouti Endong Wang Siao Xu

APPENDIX A.1. LEMMAS

Throughout the appendix we adopt the notation of Section 4. Let

$$r_{B,n} := k_B^{1.5} (v_n/n)^{(1-\mu)/2},$$

and recall the definitions of δ_n and Q_n in (4.10). As in Section 4.3, $k_A, k_B, k_u, k_\Theta \geq 1$.

LEMMA A.1 (Filtered-product concentration). *Let $\{\Phi_m^{(q)} : m \geq 0\}$, $q = 1, 2$, be deterministic linear filters satisfying $\sum_{m \geq 0} \|\Phi_m^{(q)}\|_2 = O(1)$. Under the innovation-concentration condition in Assumption 4.1(iii),*

$$\left\| \frac{1}{\sqrt{n}} \sum_{t=1}^n \sum_{a,b \geq 0} \Phi_a^{(1)} \left(u_{t-a,1} u'_{t-b,1} - \mathbf{1}\{a=b\} \Sigma_u \right) (\Phi_b^{(2)})' \right\|_{\max} = O_p(\sqrt{\tilde{v}_n}). \quad (\text{A.1.1})$$

Lemma A.1 is Lemma A.2 of Krampe, Paparoditis, and Trenkler (2023), applied under Assumption 4.1(iii).

LEMMA A.2 (Stability of the estimated companion matrix). *Under Assumption 4.1(i)–(ii), if $k_A r_{B,n} = o(1)$, then for every $\bar{\varphi} \in (\varphi, 1)$, with probability approaching one, $\hat{\mathbf{A}}$ is stable and $\|\hat{\mathbf{A}}^m\|_l \leq C k_A \bar{\varphi}^m$ for all $m \geq 0$ and $l \in \{1, \infty\}$.*

LEMMA A.3 (Consistency of the covariance estimators). *Under Assumption 4.1(i)–(iii), suppose $k_A r_{B,n} = o(1)$, $\delta_n = o(1)$, $N_n \rightarrow \infty$, and take $\tau_u \asymp \delta_n$. Then*

$$\|\hat{\Sigma}_u - \Sigma_u\|_\infty = O_p(k_u \delta_n^{1-\mu_u}), \quad (\text{A.1.2})$$

and

$$\|\hat{\Sigma}_Y - \Sigma_Y\|_\infty = O_p(k_A^2 k_u Q_n). \quad (\text{A.1.3})$$

For each receiver j ,

$$\|\hat{\Sigma}_{X,j} - \Sigma_{X,j}\|_\infty = O_p(k_A^2 k_u Q_n). \quad (\text{A.1.4})$$

LEMMA A.4 (Consistency of the nuisance estimators). *Under the conditions of Lemma A.3, suppose $k_\Theta k_A^2 k_u Q_n = o(1)$. Then, for each fixed receiver j and horizon h ,*

$$\|\hat{\Theta}_j - \Theta_j\|_\infty = O_p(k_\Theta^2 k_u k_A^2 Q_n), \quad (\text{A.1.5})$$

and

$$\max \left\{ \|\hat{\gamma}_{ij,h} - \gamma_{ij,h}\|_1, \|\hat{b}_{j,h} - b_{j,h}\|_1 \right\} = O_p(k_\Theta^2 k_u^2 k_A^5 Q_n). \quad (\text{A.1.6})$$

If, in addition, $k_\Theta^2 k_u^2 k_A^2 Q_n = o(1)$, then

$$\|\hat{\delta}_{ij,h} - \delta_{ij,h}\|_1 = O_p(k_\Theta^4 k_u^4 k_A^5 Q_n). \quad (\text{A.1.7})$$

APPENDIX A.2. PROOFS

A.2.1. Proof of Theorem 4.1

Proof. Write

$$r_{B,n} = k_B^{1.5} (v_n/n)^{(1-\mu)/2}.$$

We first collect the first-step rates used below. By Assumption 4.1(ii), $\|\hat{\mathbf{A}} - \mathbf{A}\|_l = O_p(r_{B,n})$ for $l \in \{1, \infty\}$. Fix $\bar{\varphi} \in (\varphi, 1)$. The telescoping identity

$$\hat{\mathbf{A}}^m - \mathbf{A}^m = \sum_{s=0}^{m-1} \mathbf{A}^{m-1-s} (\hat{\mathbf{A}} - \mathbf{A}) \hat{\mathbf{A}}^s$$

and Assumption 4.1(i) imply, since $k_A r_{B,n} = o(1)$,

$$\|\hat{\mathbf{A}}^m\|_l = O_p(k_A \bar{\varphi}^m), \quad \sum_{m=0}^{N_n} \|\hat{\mathbf{A}}^m - \mathbf{A}^m\|_l = O_p(k_A^2 r_{B,n}).$$

The stable companion representation $Y_t = \sum_{m \geq 0} \mathbf{A}^m J' u_{t-m-1,1}$, together with Assumption 4.1(iii), gives

$$\left\| \frac{1}{n-p} \sum_t Y_t Y_t' - \Sigma_Y \right\|_{\max} + \left\| \frac{1}{n-p} \sum_t Y_t u_{t,1}' \right\|_{\max} = O_p(\sqrt{\tilde{v}_n/n}). \quad (\text{A.2.1})$$

Moreover, $\|\Sigma_Y\|_2 = O(k_u)$ and hence $\|(n-p)^{-1} \sum_t Y_t Y_t'\|_{\max} = O_p(k_u)$.

Let $\tilde{\Sigma}_u$ denote the innovation covariance computed from the estimated one-step residuals before thresholding. Since $\hat{u}_{t,1} - u_{t,1} = -(\hat{A} - A)Y_t$,

$$\begin{aligned} \tilde{\Sigma}_u - \Sigma_u &= \frac{1}{n-p} \sum_t (u_{t,1} u_{t,1}' - \Sigma_u) \\ &\quad - (\hat{A} - A) \left(\frac{1}{n-p} \sum_t Y_t u_{t,1}' \right) - \left(\frac{1}{n-p} \sum_t u_{t,1} Y_t' \right) (\hat{A} - A)' \\ &\quad + (\hat{A} - A) \left(\frac{1}{n-p} \sum_t Y_t Y_t' \right) (\hat{A} - A)'. \end{aligned} \quad (\text{A.2.2})$$

It follows from (A.2.1) and Assumption 4.1(ii)–(iii) that

$$\|\tilde{\Sigma}_u - \Sigma_u\|_{\max} = O_p(\sqrt{\tilde{v}_n/n} + k_u r_{B,n}^2) = O_p(\delta_n).$$

The thresholding bound for $\Sigma_u \in \mathcal{U}(k_u, \mu_u)$ therefore gives

$$\|\hat{\Sigma}_u - \Sigma_u\|_\infty = O_p(k_u \delta_n^{1-\mu_u}).$$

Using the definition of $\hat{\Sigma}_Y$, the preceding bounds, and the truncation error of the stable covariance representation,

$$\begin{aligned} \|\hat{\Sigma}_Y - \Sigma_Y\|_\infty &= O_p\left(k_A^3 k_u r_{B,n} + k_A^2 k_u \delta_n^{1-\mu_u} + k_A^2 k_u \varphi^{2N_n}\right) \\ &= O_p(k_A^2 k_u Q_n). \end{aligned} \tag{A.2.3}$$

Since $\hat{\Sigma}_{X,j} = G_j \hat{\Sigma}_Y G_j'$ and $\Sigma_{X,j} = G_j \Sigma_Y G_j'$, the bounded induced norms of the grouping maps imply

$$\|\hat{\Sigma}_{X,j} - \Sigma_{X,j}\|_\infty = O_p(k_A^2 k_u Q_n). \tag{A.2.4}$$

By Assumption 4.1(v), $\|\Theta_j\|_\infty = O(k_\Theta)$. Since $k_\Theta k_A^2 k_u Q_n = o(1)$ under (4.11), the Neumann-series argument gives $\|\hat{\Theta}_j\|_\infty = O_p(k_\Theta)$. Hence

$$\begin{aligned} \|\hat{\Theta}_j - \Theta_j\|_\infty &= \left\| \hat{\Theta}_j (\Sigma_{X,j} - \hat{\Sigma}_{X,j}) \Theta_j \right\|_\infty \\ &= O_p(k_\Theta^2 k_u k_A^2 Q_n). \end{aligned} \tag{A.2.5}$$

For fixed h , $b_{j,h} = \Theta_j G_j \Sigma_Y (\mathbf{A}')^h J' e_j$ and $\hat{b}_{j,h} = \hat{\Theta}_j G_j \hat{\Sigma}_Y (\hat{\mathbf{A}}')^h J' e_j$. Using (A.2.3), (A.2.5), and the telescoping bound for $(\hat{\mathbf{A}}')^h - (\mathbf{A}')^h$ gives

$$\|\hat{b}_{j,h} - b_{j,h}\|_1 = O_p(k_\Theta^2 k_u^2 k_A^5 Q_n). \tag{A.2.6}$$

Since $\hat{\gamma}_{j,h} - \gamma_{j,h} = R_{-i,j}(\hat{b}_{j,h} - b_{j,h})$, the same rate holds for $\hat{\gamma}_{j,h} - \gamma_{j,h}$.

We now establish the linear representation. Stability and the concentration conditions imply

$$\left\| T^{-1} \sum_t X_{j,t} X'_{j,t} - \Sigma_{X,j} \right\|_{\max} = O_p(\sqrt{\tilde{v}_n/n}), \quad (\text{A.2.7})$$

$$\left\| T^{-1} \sum_t X_{j,t} u_{j,t,h} \right\|_{\max} = O_p(\sqrt{\tilde{v}_n/n}). \quad (\text{A.2.8})$$

The nonsingular factor $(R_{ij} \hat{\Theta}_j R'_{ij})^{-1}$ in $\hat{v}_{ij,t}$ cancels from (4.1). Therefore

$$\begin{aligned} \hat{\beta}_{ij,h} - \beta_{ij,h} = & \left(T^{-1} \sum_t R_{ij} \hat{\Theta}_j X_{j,t} X'_{ij,t} \right)^{-1} \left[T^{-1} \sum_t R_{ij} \hat{\Theta}_j X_{j,t} u_{j,t,h} \right. \\ & \left. + \left(T^{-1} \sum_t R_{ij} \hat{\Theta}_j X_{j,t} X'_{-i,j,t} \right) (\gamma_{j,h} - \hat{\gamma}_{ij,h}) \right]. \end{aligned} \quad (\text{A.2.9})$$

Because $\hat{\Theta}_j \hat{\Sigma}_{X,j} = I$ and $R_{ij} R'_{ij} = I_{k_{ij}}$,

$$\begin{aligned} & R_{ij} \hat{\Theta}_j \left(T^{-1} \sum_t X_{j,t} X'_{j,t} \right) R'_{ij} - I_{k_{ij}} \\ &= R_{ij} \hat{\Theta}_j \left(T^{-1} \sum_t X_{j,t} X'_{j,t} - \Sigma_{X,j} \right) R'_{ij} + R_{ij} \hat{\Theta}_j (\Sigma_{X,j} - \hat{\Sigma}_{X,j}) R'_{ij}. \end{aligned} \quad (\text{A.2.10})$$

Equations (A.2.4) and (A.2.7) yield

$$\left\| R_{ij} \hat{\Theta}_j \left(T^{-1} \sum_t X_{j,t} X'_{j,t} \right) R'_{ij} - I_{k_{ij}} \right\|_{\max} = o_p(1). \quad (\text{A.2.11})$$

Since k_{ij} is fixed,

$$\left(T^{-1} \sum_t R_{ij} \hat{\Theta}_j X_{j,t} X'_{ij,t} \right)^{-1} = I_{k_{ij}} + o_p(1).$$

Similarly, since $R_{ij} R'_{-i,j} = 0$,

$$R_{ij} \hat{\Theta}_j \left(T^{-1} \sum_t X_{j,t} X'_{j,t} \right) R'_{-i,j}$$

$$= R_{ij} \hat{\Theta}_j \left(T^{-1} \sum_t X_{j,t} X'_{j,t} - \Sigma_{X,j} \right) R'_{-i,j} + R_{ij} \hat{\Theta}_j (\Sigma_{X,j} - \hat{\Sigma}_{X,j}) R'_{-i,j}. \quad (\text{A.2.12})$$

Together with (A.2.6),

$$\begin{aligned} & \sqrt{T} \left\| \left(T^{-1} \sum_t R_{ij} \hat{\Theta}_j X_{j,t} X'_{-i,j,t} \right) (\hat{\gamma}_{j,h} - \gamma_{j,h}) \right\|_{\max} \\ &= O_p \left(\sqrt{\bar{v}_n} k_{\Theta}^3 k_u^2 k_A^5 Q_n + \sqrt{n} k_{\Theta}^3 k_u^3 k_A^7 Q_n^2 \right) = o_p(1) \end{aligned} \quad (\text{A.2.13})$$

by (4.11).

Moreover, by (A.2.5) and (A.2.8),

$$T^{-1/2} \sum_t R_{ij} (\hat{\Theta}_j - \Theta_j) X_{j,t} u_{j,t,h} = o_p(1). \quad (\text{A.2.14})$$

Substitution into (A.2.9) gives

$$\sqrt{T} (\hat{\beta}_{ij,h} - \beta_{ij,h}) = T^{-1/2} \sum_t R_{ij} \Theta_j X_{j,t} u_{j,t,h} + o_p(1). \quad (\text{A.2.15})$$

Since

$$v_{ij,t} = \Sigma_{ij \cdot -i} R_{ij} \Theta_j X_{j,t}, \quad \Sigma_{ij \cdot -i} = (R_{ij} \Theta_j R'_{ij})^{-1},$$

we obtain

$$\sqrt{T} (\hat{\beta}_{ij,h} - \beta_{ij,h}) = \Sigma_{ij \cdot -i}^{-1} T^{-1/2} \sum_t v_{ij,t} u_{j,t,h} + o_p(1), \quad (\text{A.2.16})$$

which proves (4.13). Assumption 4.1(vi) then gives

$$\sqrt{T} (\hat{\beta}_{ij,h} - \beta_{ij,h}) \xrightarrow{d} N \left(0, \Sigma_{ij \cdot -i}^{-1} \Omega_{ij,h} \Sigma_{ij \cdot -i}^{-1} \right). \quad (\text{A.2.17})$$

It remains to establish consistency of the feasible HAC estimator. First, stability implies

$\|\Sigma_{X,j}\|_2 = O(k_u)$. Since $\Sigma_{ij\cdot-i}$ is the Schur complement associated with the source block,

$$0 \preceq \Sigma_{ij\cdot-i} \preceq R_{ij}\Sigma_{X,j}R'_{ij}.$$

As k_{ij} is fixed, $\|\Sigma_{ij\cdot-i}\|_\infty = O(k_u)$.

Set $A_{ij} = R_{ij}\Theta_j R'_{ij}$ and $\hat{A}_{ij} = R_{ij}\hat{\Theta}_j R'_{ij}$. Equation (A.2.5) gives

$$\|\hat{A}_{ij} - A_{ij}\|_\infty = O_p(k_\Theta^2 k_u k_A^2 Q_n).$$

Since $k_\Theta^2 k_u^2 k_A^2 Q_n = o(1)$ under (4.11), the fixed-dimensional inverse-perturbation formula yields

$$\|\hat{\Sigma}_{ij\cdot-i} - \Sigma_{ij\cdot-i}\|_\infty = O_p(k_\Theta^2 k_u^3 k_A^2 Q_n), \quad \|\hat{\Sigma}_{ij\cdot-i}\|_\infty = O_p(k_u). \quad (\text{A.2.18})$$

Using $\hat{v}_{ij,t} = \hat{\Sigma}_{ij\cdot-i} R_{ij} \hat{\Theta}_j X_{j,t}$, $v_{ij,t} = \Sigma_{ij\cdot-i} R_{ij} \Theta_j X_{j,t}$, (A.2.5), and (A.2.18),

$$\|\hat{v}_{ij,t} - v_{ij,t}\|_2 = O_p(k_\Theta^3 k_u^3 k_A^2 Q_n) \|X_{j,t}\|_\infty. \quad (\text{A.2.19})$$

Also, $\hat{u}_{j,t,h} - u_{j,t,h} = -(\hat{b}_{j,h} - b_{j,h})' X_{j,t}$, so

$$\begin{aligned} \left(T^{-1} \sum_t |\hat{u}_{j,t,h} - u_{j,t,h}|^4 \right)^{1/4} &\leq \|\hat{b}_{j,h} - b_{j,h}\|_1 \left(T^{-1} \sum_t \|X_{j,t}\|_\infty^4 \right)^{1/4} \\ &= O_p(\tilde{v}_n^{1/2} k_\Theta^2 k_u^2 k_A^5 Q_n) = o_p(1). \end{aligned} \quad (\text{A.2.20})$$

Thus $T^{-1} \sum_t \hat{u}_{j,t,h}^4 = O_p(1)$.

Using

$$\hat{v}_{ij,t} \hat{u}_{j,t,h} - v_{ij,t} u_{j,t,h} = (\hat{v}_{ij,t} - v_{ij,t}) \hat{u}_{j,t,h} + v_{ij,t} (\hat{u}_{j,t,h} - u_{j,t,h}),$$

Cauchy–Schwarz and the fourth-moment conditions imply

$$\left(T^{-1} \sum_t \|\hat{v}_{ij,t} \hat{u}_{j,t,h} - v_{ij,t} u_{j,t,h}\|_2^2 \right)^{1/2} = O_p \left(\tilde{v}_n^{1/2} k_{\Theta}^3 k_u^3 k_A^5 Q_n \right). \quad (\text{A.2.21})$$

Let $\tilde{\Omega}_{ij,h}$ be the Newey–West estimator formed from the population score $v_{ij,t} u_{j,t,h}$. Expanding the feasible autocovariances around their population-score counterparts and applying Cauchy–Schwarz gives

$$\|\hat{\Omega}_{ij,h} - \tilde{\Omega}_{ij,h}\|_{\max} = O_p \left(b_n \tilde{v}_n^{1/2} k_{\Theta}^3 k_u^3 k_A^5 Q_n \right) = o_p(1) \quad (\text{A.2.22})$$

by (4.12). Under Assumption 4.1(vi) and the stated bandwidth conditions, $\tilde{\Omega}_{ij,h} \xrightarrow{p} \Omega_{ij,h}$, and hence

$$\hat{\Omega}_{ij,h} \xrightarrow{p} \Omega_{ij,h}.$$

Finally,

$$T^{-1} \sum_t \hat{v}_{ij,t} X'_{ij,t} = \hat{\Sigma}_{ij \cdot -i} R_{ij} \hat{\Theta}_j \left(T^{-1} \sum_t X_{j,t} X'_{j,t} \right) R'_{ij}.$$

Equations (A.2.11) and (A.2.18) imply

$$T^{-1} \sum_t \hat{v}_{ij,t} X'_{ij,t} \xrightarrow{p} \Sigma_{ij \cdot -i}.$$

Therefore

$$T \widehat{\text{AVar}}(\hat{\beta}_{ij,h}) \xrightarrow{p} \Sigma_{ij \cdot -i}^{-1} \Omega_{ij,h} \Sigma_{ij \cdot -i}^{-1}. \quad (\text{A.2.23})$$

The Wald result follows from (A.2.17) and Slutsky’s theorem. □

A.2.2. Proof of Theorem 4.2

Proof. For brevity, write $\Sigma = \Sigma_{ij \cdot -i}$, $\hat{\Sigma} = \hat{\Sigma}_{ij \cdot -i}$, $\beta = \beta_{ij,h}$, and $\hat{\beta} = \hat{\beta}_{ij,h}$.

The proof of Theorem 4.1 establishes

$$\|\hat{\Sigma} - \Sigma\|_\infty = O_p(k_\Theta^2 k_u^3 k_A^2 Q_n), \quad \|\hat{b}_{j,h} - b_{j,h}\|_1 = O_p(k_\Theta^2 k_u^2 k_A^5 Q_n). \quad (\text{A.2.24})$$

Since $\Sigma^{-1} = R_{ij} \Theta_j R'_{ij}$ and k_{ij} is fixed, $\|\Sigma^{-1}\|_2 = O(k_\Theta)$. Hence

$$\left\| \Sigma^{-1/2} (\hat{\Sigma} - \Sigma) \Sigma^{-1/2} \right\|_2 = O_p(k_\Theta^3 k_u^3 k_A^2 Q_n) = o_p(1) \quad (\text{A.2.25})$$

by (4.15).

Let $q = \beta' \Sigma \beta$ and $\hat{q} = \hat{\beta}' \hat{\Sigma} \hat{\beta}$. Equation (A.2.25) implies

$$|\beta' (\hat{\Sigma} - \Sigma) \beta| \leq q \left\| \Sigma^{-1/2} (\hat{\Sigma} - \Sigma) \Sigma^{-1/2} \right\|_2 = o_p(q). \quad (\text{A.2.26})$$

We next establish consistency of the full-projection residual variance. Let $d_{b,h} = \hat{b}_{j,h} - b_{j,h}$. Since $\hat{u}_{j,t,h} = u_{j,t,h} - d'_{b,h} X_{j,t}$,

$$\begin{aligned} \hat{\sigma}_j^2(h) - T^{-1} \sum_t u_{j,t,h}^2 &= -2d'_{b,h} \left(T^{-1} \sum_t X_{j,t} u_{j,t,h} \right) \\ &\quad + d'_{b,h} \left(T^{-1} \sum_t X_{j,t} X'_{j,t} \right) d_{b,h}. \end{aligned} \quad (\text{A.2.27})$$

By Assumption 4.1(iv), (A.2.7), and (A.2.24), the two terms on the right-hand side are, respectively,

$$O_p(k_\Theta^2 k_u^2 k_A^5 Q_n \sqrt{\tilde{v}_n/n}) \quad \text{and} \quad O_p(k_\Theta^4 k_u^5 k_A^{10} Q_n^2),$$

both of which are $o_p(1)$ under (4.15). Assumption 4.1(vi) therefore gives

$$\hat{\sigma}_j^2(h) \xrightarrow{p} \sigma_j^2(h). \quad (\text{A.2.28})$$

By Assumption 4.1(vi), $\sigma_j^2(h) \geq c > 0$.

It remains to account for estimation of β . From (4.13) and the score CLT,

$$\|\hat{\beta} - \beta\|_2 = O_p(k_\Theta/\sqrt{T}).$$

Moreover, $\|\hat{\Sigma}\|_2 = O_p(k_u)$ by (A.2.18). Hence

$$r_{\beta,n} := (\hat{\beta} - \beta)' \hat{\Sigma} (\hat{\beta} - \beta) = O_p(k_u k_\Theta^2/T) = o_p(1). \quad (\text{A.2.29})$$

The last equality follows from (4.15), since $Q_n \geq \delta_n^{1-\mu_u} \geq n^{-1/2}$ for all sufficiently large n .

Expanding $\hat{q} - q$ gives

$$\hat{q} - q = \beta' (\hat{\Sigma} - \Sigma) \beta + 2(\hat{\beta} - \beta)' \hat{\Sigma} \beta + r_{\beta,n}.$$

By Cauchy–Schwarz in the quadratic norm induced by $\hat{\Sigma}$,

$$\left| (\hat{\beta} - \beta)' \hat{\Sigma} \beta \right| \leq r_{\beta,n}^{1/2} (\beta' \hat{\Sigma} \beta)^{1/2}.$$

Since $\beta' \hat{\Sigma} \beta = q\{1 + o_p(1)\}$ by (A.2.25), we obtain

$$\begin{aligned} \frac{|\hat{q} - q|}{1+q} &\leq \frac{|\beta' (\hat{\Sigma} - \Sigma) \beta|}{1+q} + 2 \frac{r_{\beta,n}^{1/2} (\beta' \hat{\Sigma} \beta)^{1/2}}{1+q} + \frac{r_{\beta,n}}{1+q} \\ &= o_p(1), \end{aligned} \quad (\text{A.2.30})$$

where $\sqrt{q}/(1+q) \leq 1/2$.

Finally,

$$s_{ij}(h) = \log \left(\frac{\sigma_j^2(h) + q}{\sigma_j^2(h)} \right), \quad \hat{s}_{ij}(h) = \log \left(\frac{\hat{\sigma}_j^2(h) + \hat{q}}{\hat{\sigma}_j^2(h)} \right).$$

Since $\sigma_j^2(h) \geq c > 0$,

$$\sigma_j^2(h) + q \geq \min\{c, 1\}(1+q).$$

Equations (A.2.28) and (A.2.30) therefore imply

$$\frac{\hat{\sigma}_j^2(h) + \hat{q}}{\sigma_j^2(h) + q} \xrightarrow{p} 1, \quad \frac{\hat{\sigma}_j^2(h)}{\sigma_j^2(h)} \xrightarrow{p} 1.$$

Thus $\hat{s}_{ij}(h) - s_{ij}(h) \xrightarrow{p} 0$.

□